

自然语言处理

WELCOME!

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基于Yulia, Junjie, Diyi等课件更新

面试问题 (Part I)

- 请用 1 句话说明：为什么直接把词写成字符串(如 “苹果” “Orange”) 无法让计算机理解这些语义关系？
- “假设你是一个搜索引擎工程师，需要设计 ‘关键词匹配’ 功能。用户搜索 ‘机器学习入门教程’，你有下述两篇文档。根据TF-IDF思想，你会优先推荐哪篇文档：
 - 文档 A：包含 10 次 ‘机器学习’，1 次 ‘入门’，0 次 ‘教程’；
 - 文档 B：包含 3 次 ‘机器学习’，2 次 ‘入门’，2 次 ‘教程’
- LSA跟TF-IDF相比的优劣势是什么？

面试问题 (Part II)

- 常用的词向量学习有哪些方法？它们有什么不同和优劣势？
- 如何评估一个词向量表示的优劣？

大纲

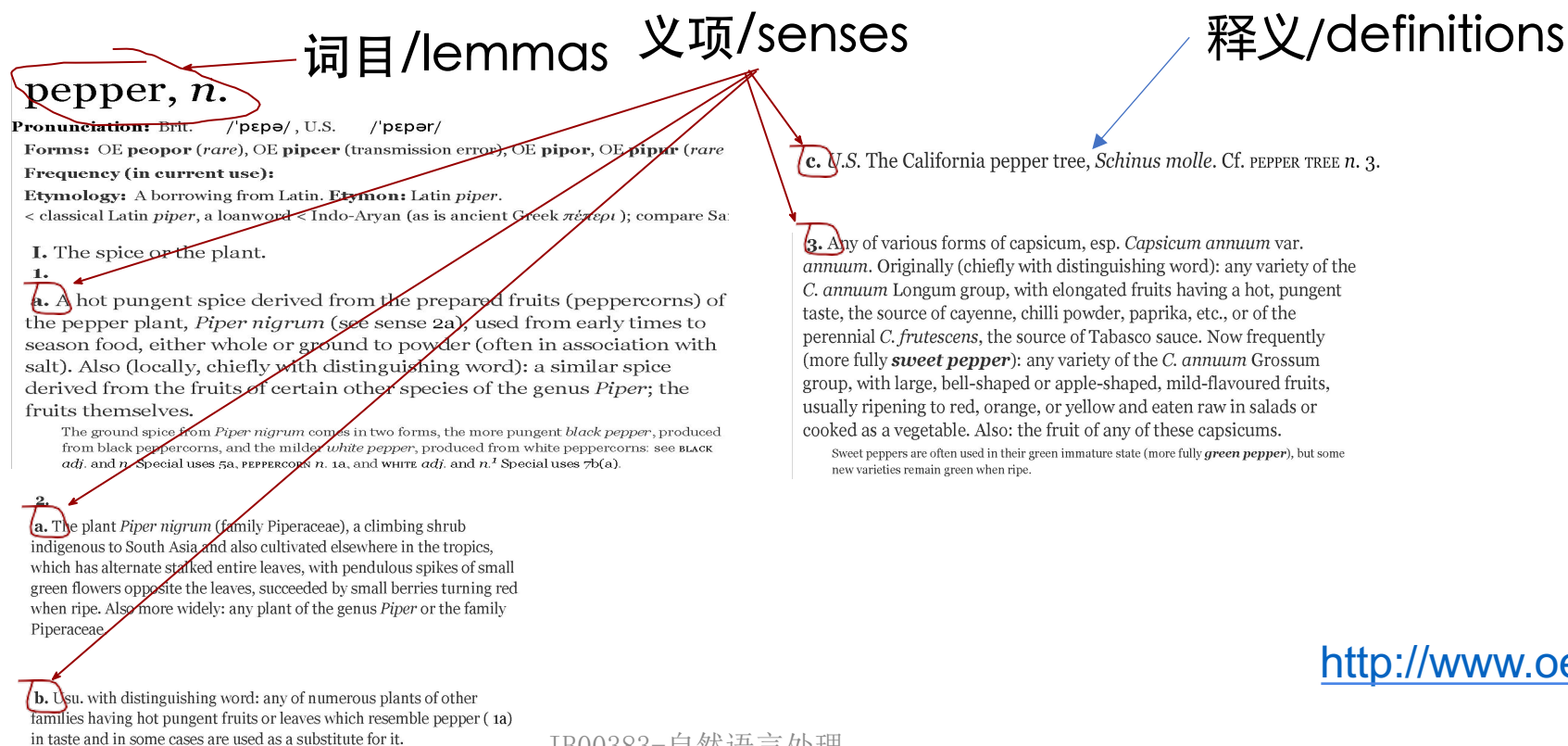
- 词汇语义学
- 电子词典
- 词的离散表示
- 词向量
- 词表示的评估

词汇语义学/Lexical Semantics: What do Words Mean?

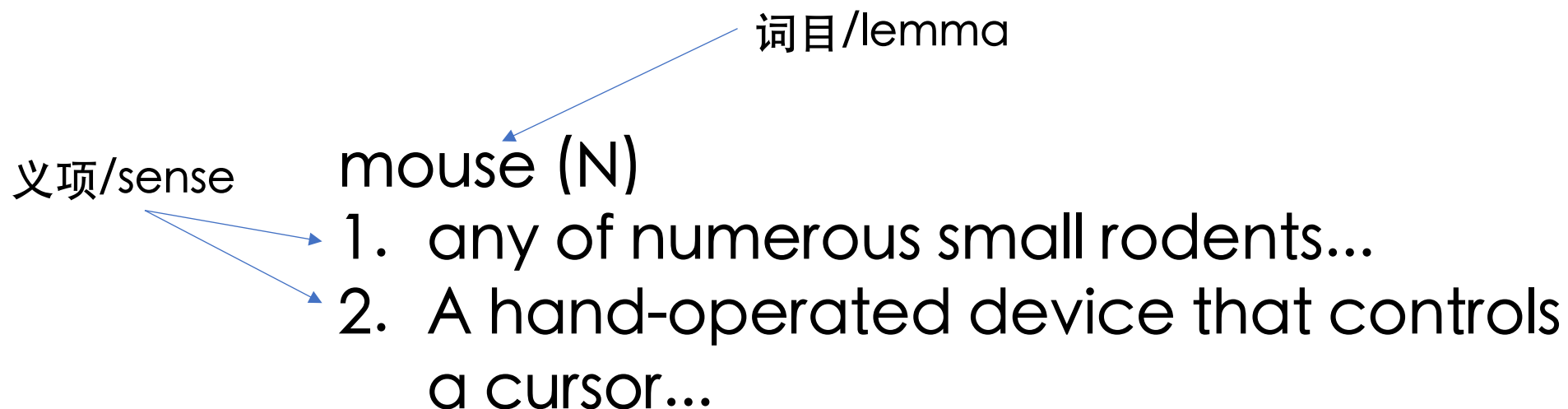
- 语言模型的N-gram或文本分类方法
 - 单词只是字符串（或词汇表中的索引 w_i ）
- 迫切需要之物/Desiderata

词汇语义学/Lexical Semantics

- 我们应该如何表示这个词的意思？
 - 词/words，词目/lemmas，义项/senses，释义/definitions



词元和词义/Lemmas and Senses



- 义项/sense或“概念/concept”是一个词的含义组成部分/meaning component
- 词目/lemmas可以是多义的/polysemous (有多个义项/senses)

词间关系

- 同义性/Synonymity
- 反义词/Antonymy
- 相似性/Similarity
- 词相关性/Word Relatedness、词关联/Word Association
- 上位/下位(从属) Superordinate/Subordinate
- 整体 - 部分关系 Holonymy-Meronymy
- 因果关系 Causality
- 同音/同形异义关系 Homonymy (Homophony/Homography)
- 多义关系 Polysemy

关系：同义性/Synonymity

- 同义词/Synonyms在某些或所有上下文中具有相同的含义
 - filbert / hazelnut
 - couch / sofa
 - big / large
 - automobile / car
 - vomit / throw up
 - water / H₂O
 - 看 / 瞧 / 瞥

语言学对比原则

The Linguistic Principle of Contrast

Difference in form → difference in meaning

- **注意:** 可能不存在完全的同义词
 - 即使意思的许多方面是相同的
 - 基于礼貌/politeness、俚语/slang、语域/register、体裁/genre等概念，同义词仍然可能无法保持可接受性
 - Water / H₂O in a surfing guide?
 - my big sister != my large sister

关系：反义词/Antonymy

- 义项/Senses在某个特征方面是相反的/opposite
 - 其他方面都很相似
 - dark/light short/long fast/slow rise/fall
 - hot/cold up/down in/out
- 更正式的说法是：反义词
 - 定义二元对立或处于尺度的两端：
 - long/short 长/短, fast/slow 快/慢
 - 可以反映逆反关系：
 - rise/fall 上升/下降; up/down 上/下

关系：相似性/Similarity

- 意思相近的词
 - 不是同义词，而是共享部分语义要素
 - Car, bicycle
 - Cow, horse

- 标注：两个词意思有多相近

word1	word2	similarity
vanish	disappear	9.8
behave	obey	7.3
belief	impression	5.95
muscle	bone	3.65
modest	flexible	0.98
hole	agreement	0.3

Felix Hill, Roi Reichart, Anna Korhonen: [SimLex-999: Evaluating Semantic Models With \(Genuine\) Similarity Estimation](#). Comput. Linguistics 41(4): 665-695 (2015)

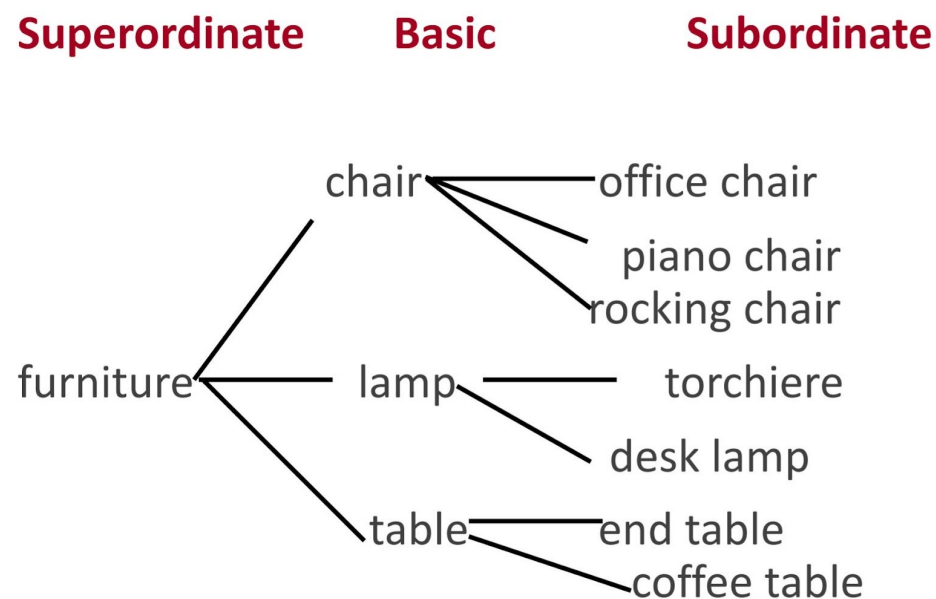
关系：词相关性/Word Relatedness

- 亦称 “词关联/word association”
- 词语可以以任何方式联系起来，可能是通过语义框架或语义场 semantic frame or field
 - car, bicycle:
 - similar
 - car, gasoline:
 - related, not similar
- 语义场/Semantic field
 - 词语
 - 覆盖特定的语义域/semantic domain
 - 相互之间保持有组织的关系
 - 例子
 - hospitals
 - surgeon, scalpel, nurse, anaesthetic, hospital
 - restaurants
 - waiter, menu, plate, food, menu, chef
 - houses
 - door, roof, kitchen, family, bed

关系：上位/下位(从属)

Superordinate/Subordinate

- 若一个义项是另一义项的**子类(更具体)**，则称前者为后者的**下义词(hyponym)**
 - car is a subordinate of vehicle
 - mango is a subordinate of fruit
- 反之，则称后者为前者的**上义词(hyponym)**
 - vehicle is a superordinate of car
 - fruit is a superordinate of mango
- 分类学/Taxonomy
 - 域界门纲目科属种



词汇语义学/Lexical Semantics

- 应该如何表征一个词的意思？
 - 词典释义 (Dictionary definition)
 - 词目与词形 (Lemma and wordforms)
 - 义项 (Senses)
 - 词与词或义项之间的语义关系 (Relationships between words or senses)
 - 分类学关系 (Taxonomic relationships)
 - 词语相似性与词语相关性 (Word similarity, word relatedness)
 - 语义框架与语义角色 (Semantic frames and roles)
 - 内涵义与情感倾向 (Connotation and sentiment)

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 - *John hit Bill*
 - *Bill was hit by John*

词汇语义学/Lexical Semantics

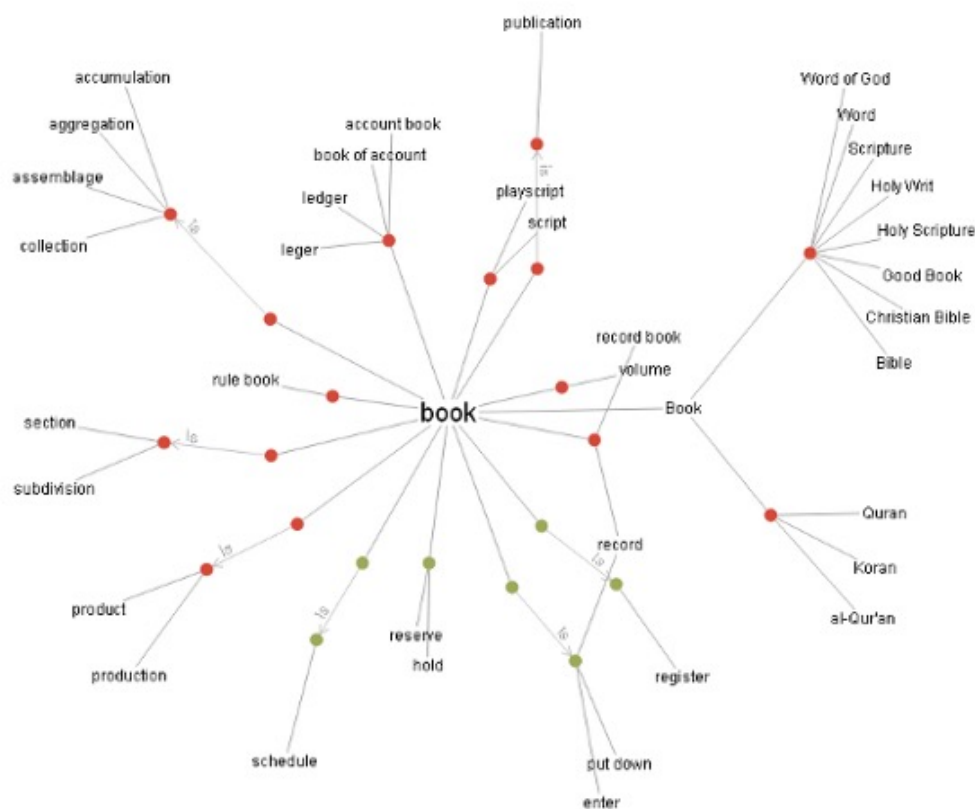
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 - 效价(valence): 刺激的愉悦程度
 - 唤醒度(arousal): 情绪激发的强度
 - 支配度(dominance): 刺激所施加的控制或支配程度

	Valence	Arousal	Dominance
courageous	8.05	5.5	7.38
music	7.67	5.57	6.5
heartbreak	2.45	5.65	3.58
cub	6.71	3.95	4.24
life	6.68	5.59	5.89

电子词典 Electronic Dictionaries

- WordNet

- <https://wordnet.princeton.edu/>



WordNet Search - 3.1
- [WordNet home page](#) - [Glossary](#) - [Help](#)

Word to search for:

Display Options:

Key: "S:" = Show Synset (semantic) relations, "W:" = Show Word (lexical) relations
Display options for sense: (gloss) "an example sentence"

Noun

- **S: (n) bank** (sloping land (especially the slope beside a body of water)) "they pulled the canoe up on the bank"; "he sat on the bank of the river and watched the currents"
- **S: (n) depository financial institution, bank, banking concern, banking company** (a financial institution that accepts deposits and channels the money into lending activities) "he cashed a check at the bank"; "that bank holds the mortgage on my home"
- **S: (n) bank** (a long ridge or pile) "a huge bank of earth"
- **S: (n) bank** (an arrangement of similar objects in a row or in tiers) "he operated a bank of switches"
- **S: (n) bank** (a supply or stock held in reserve for future use (especially in emergencies))
- **S: (n) bank** (the funds held by a gambling house or the dealer in some gambling games) "he tried to break the bank at Monte Carlo"
- **S: (n) bank, cant, camber** (a slope in the turn of a road or track; the outside is higher than the inside in order to reduce the effects of centrifugal force)

电子词典

Electronic Dictionaries

- WordNet

- NLTK: www.nltk.org

```
from nltk.corpus import wordnet as wn
panda = wn.synset('panda.n.01')
hyper = lambda s: s.hypernyms()
list(panda.closure(hyper))
```

```
[Synset('procyonid.n.01'),
Synset('carnivore.n.01'),
Synset('placental.n.01'),
Synset('mammal.n.01'),
Synset('vertebrate.n.01'),
Synset('chordate.n.01'),
Synset('animal.n.01'),
Synset('organism.n.01'),
Synset('living_thing.n.01'),
Synset('whole.n.02'),
Synset('object.n.01'),
Synset('physical_entity.n.01'),
Synset('entity.n.01')]
```

离散表示的问题

- 细度太粗
 - *expert* ↔ *skillful*
- 稀疏的
 - *wicked, badass, ninja*
- 主观的
- 昂贵的
- 很难计算词关系

S: (adj) full, good
S: (adj) estimable, good, honorable, respectable
S: (adj) beneficial, good
S: (adj) good, just, upright
S: (adj) adept, expert, good, practiced, proficient, skillful
S: (adj) dear, good, near
S: (adj) good, right, ripe
...
S: (adv) well, good
S: (adv) thoroughly, soundly, good
S: (n) good, goodness
S: (n) commodity, trade good, good

expert [0 0 0 **1** 0 0 0 0 0 0 0 0 0 0 0 0]

skillful [0 0 0 0 0 0 0 0 0 0 0 **1** 0 0 0 0]

- 高维度: PTB: 50K, Google1T: 13M

分布假说 Distributional Hypothesis

词义即其用法

“The meaning of a word is its use in the language”

[Wittgenstein PI 43]

观其伴侣，而知其义

“You shall know a word by the company it keeps”

[Firth 1957]

分布相似的词，语义相近

If A and B have almost identical environments we say that they are synonyms.

[Harris 1954]

物以类聚，词以群分

例子

- Ongchoi是什么意思？
- 假设你看到这样的描述
 - Ongchoi is delicious sautéed with garlic
 - Ongchoi is superb over rice
 - Ongchoi leaves with salty sauces
- 你又见过下面的描述
 - ...spinach sautéed with garlic over rice
 - Chard stems and leaves are delicious
 - Collard greens and other salty leafy greens
- 空心菜炒蒜蓉超好吃
- 空心菜配米饭绝了
- 空心菜叶配咸酱最棒
- 菠菜炒蒜蓉配白饭
- 甜菜梗和叶子都很好吃
- 羽衣甘蓝和其他咸味叶菜

Ongchoi: Ipomoea aquatica "Water Spinach"

- Ongchoi is a leafy green like spinach, chard, or collard greens
- 空心菜是一种绿叶蔬菜，像菠菜、甜菜或羽衣甘蓝

空心菜
Kangkong
Rau Muong
...



https://commons.wikimedia.org/wiki/File:Ong_choy_water_spinach.png

词嵌入/Word Embedding

- 把词的意思定义为一个向量
 - 嵌入/embeddings: 将词嵌入某个空间
- 每个现代NLP算法都使用嵌入作为词含义的表示



直觉：为什么使用向量？

- 以情感分析/sentiment analysis任务为例：
 - 用词作为特征，该特征是词本身
 - 构建某个特征：前一个词是“terrible”
 - 需要：训练和测试集中词完全匹配
 - 若使用嵌入/embeddings：
 - 特征是向量
 - 如上例：前一个词是向量[35, 22, 17]
 - 测试集中可能看到类似的向量[35, 22, 14]
 - 可以泛化到相似但没有见过的词!!!

嵌入有很多类型

- 基于计数(Count-based)的方法
 - 单词由附近单词的计数的简单函数表示
- 基于类(Class-based)的方法
 - 表示是通过分层聚类，布朗聚类创建的
- 分布式基于预测的(类型)嵌入
 - 词表示是通过训练分类器来区分附近和远处的单词来创建的：word2vec，fasttext
- 语言模型的分布式上下文(词元)嵌入
 - ELMo, BERT

先讨论两种词嵌入

- TF-IDF
 - 信息检索主力/workhorse !
 - 基线模型
 - 稀疏向量/Sparse vectors
 - 单词由附近单词的计数（一个简单的函数）表示
- Word2vec
 - 稠密向量/Dense vectors
 - 表示是通过训练分类器来预测一个词是否可能出现在附近来创建的
 - <https://fasttext.cc/docs/en/crawl-vectors.html>
 - 稍后我们将讨论称为上下文嵌入的扩展

词-文档矩阵/Term-document Matrix

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	17
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

- 上下文/Context=出现在同一文档中

词-文档矩阵/Term-document Matrix

	As You Like It	Twelfth Night	Julius Caesar	Henry V
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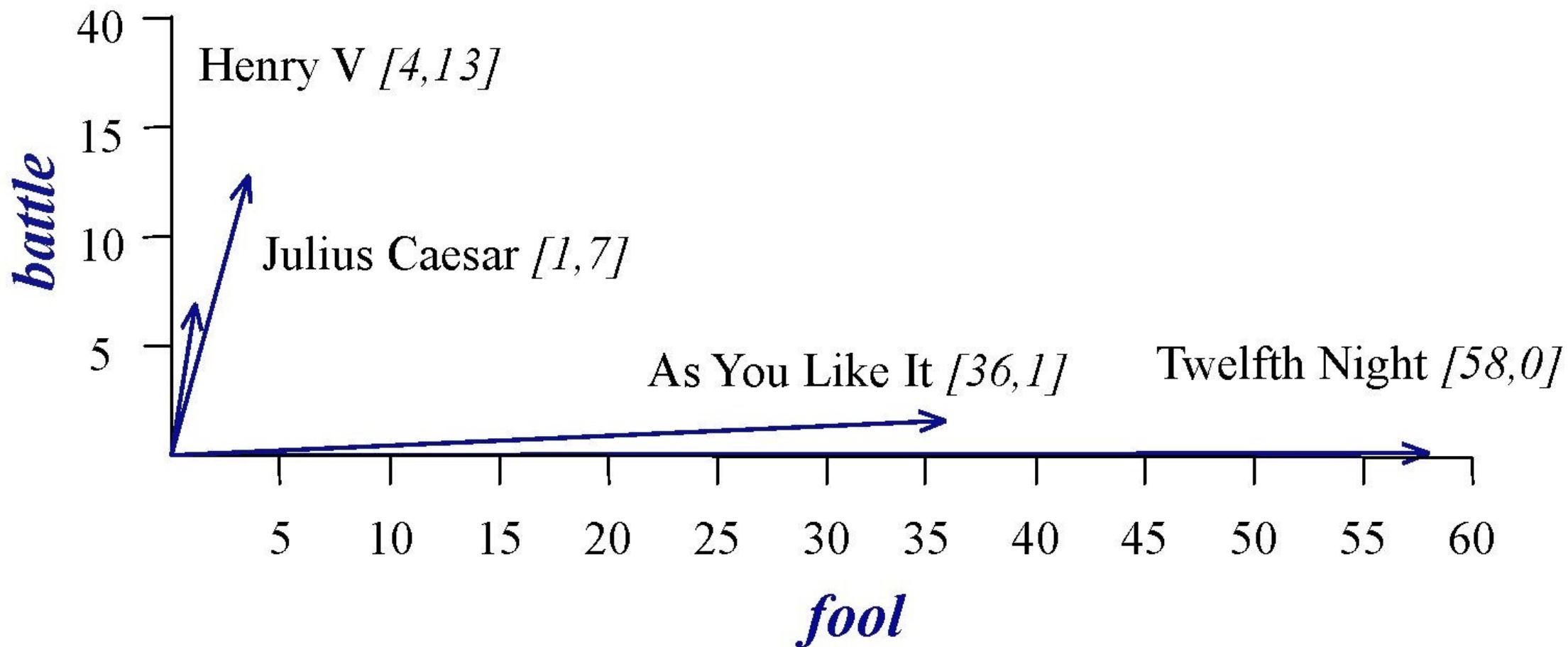
- 两部喜剧/comedies的向量是相似的,与历史剧不同
- 喜剧中fool、wit出现更多，更少battle

向量是信息检索的基础

	As You Like It	Twelfth Night	Julius Caesar	Henry V
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- 两部喜剧/comedies的向量是相似的,与历史剧不同
- 喜剧中fool、wit出现更多，更少battle

文档向量可视化



词也可以是向量

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	17
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

- battle 更多出现在Julius Caesar和Henry V两部剧中
- Fool更多出现在喜剧里的词，特别是Twelfth Night

常见设置: 词-词矩阵 (词-上下文矩阵 /Term-Context Matrix)

	knife	dog	sword	love	like
knife	0	1	6	5	5
dog	1	0	5	5	5
sword	6	5	0	5	5
love	5	5	5	0	5
like	5	5	5	5	2

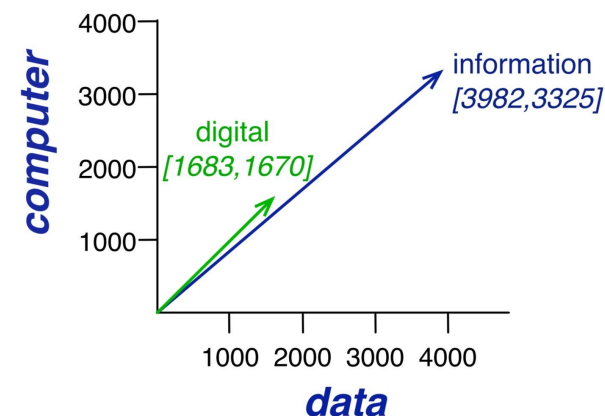
- 如果两个词的上下文向量相似，那么它们的意思就“相似”
 - 相似性=相关性

词-上下文矩阵/Term-Context Matrix

- 两个词的上下文向量相似，那么它们的意思就“相似”

is traditionally followed by **cherry** pie, a traditional dessert
often mixed, such as **strawberry** rhubarb pie. Apple pie
computer peripherals and personal **digital** assistants. These devices usually
a computer. This includes **information** available on the internet

	aardvark	...	computer	data	result	pie	sugar	...
cherry	0	...	2	8	9	442	25	...
strawberry	0	...	0	0	1	60	19	...
digital	0	...	1670	1683	85	5	4	...
information	0	...	3325	3982	378	5	13	...



计算词语相似度

Computing Word Similarity

- 两个向量的内积/点乘(dot-product)是一个标量:

$$\mathbf{v}^T \mathbf{w} = \mathbf{v} \cdot \mathbf{w} = \sum_{i=1}^d v_i w_i = v_1 w_1 + v_2 w_2 + \cdots + v_d w_d$$

- 当两个向量在相同的维数中有较大的值时，内积更大
- 因此，内积/点乘可以作为向量之间有用的相似性度量

未处理的点乘的问题

- 点乘倾向于长向量
 - 如果一个向量较长(在许多维度上具有较高的值)，则点乘值较高

$$\|\mathbf{v}\| = \sqrt{\sum_{i=1}^d v_i^2}$$

- 高频词(如of、the、you)有很长的向量(因为它们与其他单词一起出现很多次)
 - 所以点乘过于倾向于使用高频词

备选:用余弦计算单词相似度

$$\cos(\mathbf{v}, \mathbf{w}) = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \|\mathbf{w}\|} = \frac{\sum_{i=1}^d v_i w_i}{\sqrt{\sum_{i=1}^d v_i^2} \sqrt{\sum_{i=1}^d w_i^2}}$$

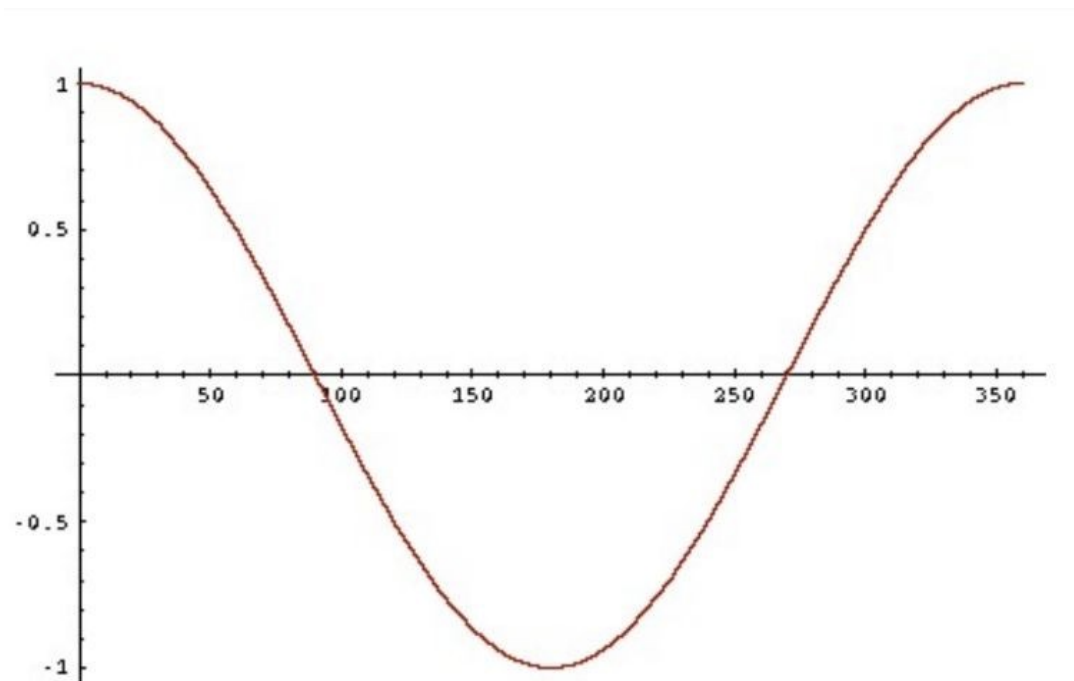
- 由两个向量 $\mathbf{a}, \mathbf{b}$ 的定义，说明

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos\theta, \quad \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} = \cos\theta$$

- 意义:

- -1: 两个向量反向
- +1: 两个向量同向
- 0: 两个向量垂直

- 注意: 由于词频是非负，所以词-词矩阵相似度的范围0-1



In-class Practice I

基于计数的表示

Count-based Representations

	As You Like It	Twelfth Night	Julius Caesar	Henry V
battle	1	0	7	17
good	114	80	62	89
fool	36	58	1	4
wit	20	15	2	3

- 计数/Counts: 词频/term-frequency
 - 移除停用词
 - 使用 $\log_{10}(\text{tf})$
 - 用文档长度归一化

原始词频是个坏表示

- 共现/co-occurrence矩阵用词频表示每个单元格
- 词频率显然有用；如果sugar(糖)经常跟apricot(杏)同时出现，这是有用信息
- 但是像一些高频词，如the， it， 对上下文没有带来太多信息
- 这是一个悖论！
 - 我们如何平衡这两个相互冲突的约束？

两个常见的单词加权解决方案

- **TF-IDF**: 赋予单词 t 在文档 d 中的价值

$$w_{t,d} = \text{tf}_{t,d} \times \text{idf}_t$$

- 单词比如the, it的idf值较低

- **PMI(Pointwise Mutual Information)**: 逐点互信息(两个词的关联度)

$$\text{PMI}(\mathbf{w}_1, \mathbf{w}_2) = \log \frac{P(\mathbf{w}_1, \mathbf{w}_2)}{P(\mathbf{w}_1)P(\mathbf{w}_2)}$$

- 像“good”这样的单词与“great”一起出现的频率是否比我们预期的要高

TF-IDF

- 如何处理均匀分布在许多文档中的单词?

$$\text{tf}_{t,d} = \begin{cases} 1 + \log_{10}(\text{count}(t, d)) & \text{if } \text{count}(t, d) > 0 \\ 0 & \text{其他} \end{cases} \quad \text{或 } \log_{10}(\text{count}(t, d) + 1)$$

$$\text{idf}_t = 1 + \log_{10} \left(\frac{N}{\text{df}_t} \right)$$

集合的文档的总数

包含词 t 的文档的数目

- 像 “the” 或 “good” 这样的词的idf很低

$$w_{t,d} = \text{tf}_{t,d} \times \text{idf}_t$$

正点互信息

Positive Pointwise Mutual Information (PPMI)

- 在词-上下文(word-context)矩阵中
- 如何评估: 词 w 和 c 一起出现的次数比它们独立出现的次数多?

$$\text{PMI}(w, c) = \log_2 \frac{P(w, c)}{P(w)P(c)}$$

$$\text{PPMI}(w, c) = \max \left(\log_2 \frac{P(w, c)}{P(w)P(c)}, 0 \right)$$

- PMI偏向于低频事件
 - 非常罕见的单词具有非常高的PMI值
 - 给稀有词更高的概率 $\alpha = 0.75$

$$\text{PPMI}_\alpha(w, c) = \max \left(\log_2 \frac{P(w, c)}{P(w)P_\alpha(c)}, 0 \right), P_\alpha(c) = \frac{\text{count}(c)^\alpha}{\sum_{c'} \text{count}(c')^\alpha}$$

In-class Practice II

#	name	formula	reference
1.	Joint probability	$p(xy)$	(Giuliano, 1964)
2.	Conditional probability	$p(y x)$	(Gregory et al., 1999)
3.	Reverse cond. probability	$p(x y)$	(Gregory et al., 1999)
4.	Pointwise mutual inf. (MI)	$\log \frac{p(xy)}{p(x*)p(*y)}$	(Church and Hanks, 1990)
5.	Mutual dependency (MD)	$\log \frac{p(xy)^2}{p(x*)p(*y)}$	(Thanopoulos et al., 2002)
6.	Log frequency biased MD	$\log \frac{p(xy)^2}{p(x*)p(*y)} + \log p(xy)$	(Thanopoulos et al., 2002)
7.	Normalized expectation	$\frac{2f(xy)}{f(x*)+f(*y)}$	(Smadja and McKeown, 1990)
8.	Mutual expectation	$\frac{2f(xy)}{f(x*)+f(*y)} \cdot p(xy)$	(Dias et al., 2000)
9.	Salience	$\log \frac{p(xy)^2}{p(x*)p(*y)} \cdot \log f(xy)$	(Kilgarrieff and Tugwell, 2001)
10.	Pearson's χ^2 test	$\sum_{i,j} \frac{(f_{ij}-\hat{f}_{ij})^2}{\hat{f}_{ij}}$	(Manning and Schütze, 1999)
11.	Fisher's exact test	$\frac{f(x*)!f(\bar{x}*)!f(*y)!f(*\bar{y})!}{N!f(xy)!f(x\bar{y})!f(\bar{x}y)!f(\bar{x}\bar{y})!}$	(Pedersen, 1996)
12.	t test	$\frac{f(xy)-\hat{f}(xy)}{\sqrt{f(xy)(1-(f(xy)/N))}}$	(Church and Hanks, 1990)
13.	z score	$\frac{f(xy)-\hat{f}(xy)}{\sqrt{\hat{f}(xy)(1-(\hat{f}(xy)/N))}}$	(Berry-Rogghe, 1973)
14.	Poisson significance	$\frac{f(xy)-f(xy)\log f(xy)+\log f(xy)!}{\log N}$	(Quasthoff and Wolff, 2002)
15.	Log likelihood ratio	$-2 \sum_{i,j} f_{ij} \log \frac{f_{ij}}{\hat{f}_{ij}}$	(Dunning, 1993)
16.	Squared log likelihood ratio	$-2 \sum_{i,j} \frac{\log f_{ij}^2}{\hat{f}_{ij}}$	(Inkpen and Hirst, 2002)
17.	Russel-Rao	$\frac{a}{a+b+c+d}$	(Russel and Rao, 1940)
18.	Sokal-Michiner	$\frac{a+d}{a+b+c+d}$	(Sokal and Michener, 1958)
19.	Rogers-Tanimoto	$\frac{a+d}{a+2b+2c+d}$	(Rogers and Tanimoto, 1960)
20.	Hamann	$\frac{(a+d)-(b+c)}{a+b+c+d}$	(Hamann, 1961)
21.	Third Sokal-Sneath	$\frac{b+c}{a+d}$	(Sokal and Sneath, 1963)
22.	Jaccard	$\frac{a}{a+b+c}$	(Jaccard, 1912)
23.	First Kulczynsky	$\frac{a}{b+c}$	(Kulczynski, 1927)
24.	Second Sokal-Sneath	$\frac{a}{a+2(b+c)}$	(Sokal and Sneath, 1963)
25.	Second Kulczynski	$\frac{1}{2}(\frac{a}{a+b} + \frac{a}{a+c})$	(Kulczynski, 1927)
26.	Fourth Sokal-Sneath	$\frac{1}{4}(\frac{a}{a+b} + \frac{a}{a+c} + \frac{d}{d+b} + \frac{d}{d+c})$	(Kulczynski, 1927)
27.	Odds ratio	$\frac{ad}{bc}$	(Tan et al., 2002)
28.	Yulle's ω	$\frac{\sqrt{ad}-\sqrt{bc}}{\sqrt{ad}+\sqrt{bc}}$	(Tan et al., 2002)
29.	Yulle's Q	$\frac{ad-bc}{ad+bc}$	(Tan et al., 2002)
30.	Driver-Kroeber	$\frac{a}{\sqrt{(a+b)(a+c)}}$	(Driver and Kroeber, 1932)

#	name	formula	reference
31.	Fifth Sokal-Sneath	$\frac{ad}{\sqrt{(a+b)(a+c)(d+b)(d+c)}}$	(Sokal and Sneath, 1963)
32.	Pearson	$\frac{ad-bc}{\sqrt{(a+b)(a+c)(d+b)(d+c)}}$	(Pearson:1950)
33.	Baroni-Urbani	$\frac{a+\sqrt{ad}}{a+b+c+\sqrt{ad}}$	(Baroni-Urbani and Buser, 1976)
34.	Braun-Blanquet	$\frac{a}{\max(a+b,a+c)}$	(Braun-Blanquet, 1932)
35.	Simpson	$\frac{a}{\min(a+b,a+c)}$	(Simpson, 1943)
36.	Michael	$\frac{4(ad-bc)}{(a+d)^2+(b+c)^2}$	(Michael, 1920)
37.	Mountford	$\frac{2a}{2bc+ab+ac}$	(Kaufman and Rousseeuw, 1990)
38.	Fager	$\frac{a}{\sqrt{(a+b)(a+c)}} - \frac{1}{2} \max(b, c)$	(Kaufman and Rousseeuw, 1990)
39.	Unigram subtuples	$\log \frac{ad}{bc} - 3.29\sqrt{\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}}$	(Blaheta and Johnson, 2001)
40.	U cost	$\log(1 + \frac{\min(b,c)+a}{\max(b,c)+a})$	(Tulloss, 1997)
41.	S cost	$\log(1 + \frac{\min(b,c)}{a+1})^{-\frac{1}{2}}$	(Tulloss, 1997)
42.	R cost	$\log(1 + \frac{a}{a+b}) \cdot \log(1 + \frac{a}{a+c})$	(Tulloss, 1997)
43.	T combined cost	$\sqrt{U \times S \times R}$	(Tulloss, 1997)
44.	Phi	$\frac{p(xy)-p(x*)p(*y)}{\sqrt{p(x*)p(*y)(1-p(x*))(1-p(*y))}}$	(Tan et al., 2002)
45.	Kappa	$\frac{p(xy)+p(\bar{x}\bar{y})-p(x*)p(*y)-p(\bar{x}*)p(*\bar{y})}{1-p(x*)p(*y)-p(\bar{x}*)p(*\bar{y})}$	(Tan et al., 2002)
46.	J measure	$\max[p(xy) \log \frac{p(y x)}{p(*y)} + p(x\bar{y}) \log \frac{p(\bar{y} \bar{x})}{p(*\bar{y})},$ $p(xy) \log \frac{p(x y)}{p(x*)} + p(\bar{x}y) \log \frac{p(\bar{x} \bar{y})}{p(\bar{x}*)}]$	(Tan et al., 2002)
47.	Gini index	$\max[p(x*)(p(y x)^2 + p(\bar{y} \bar{x})^2) - p(*y)^2$ $+ p(\bar{x}*)(p(y \bar{x})^2 + p(\bar{y} \bar{x})^2) - p(*\bar{y})^2,$ $p(*y)(p(x y)^2 + p(\bar{x} \bar{y})^2) - p(x*)^2$ $+ p(*\bar{y})(p(x \bar{y})^2 + p(\bar{x} \bar{y})^2) - p(\bar{x}*)^2]$	(Tan et al., 2002)
48.	Confidence	$\max[p(y x), p(x y)]$	(Tan et al., 2002)
49.	Laplace	$\max[\frac{Np(xy)+1}{Np(x*)+2}, \frac{Np(xy)+1}{Np(*y)+2}]$	(Tan et al., 2002)
50.	Conviction	$\max[\frac{p(x*)p(*y)}{p(x\bar{y})}, \frac{p(\bar{x}*)p(*y)}{p(\bar{x}y)}]$	(Tan et al., 2002)
51.	Piatersky-Shapiro	$p(xy) - p(x*)p(*y)$	(Tan et al., 2002)
52.	Certainty factor	$\max[\frac{p(y x)-p(*y)}{1-p(*y)}, \frac{p(x y)-p(x*)}{1-p(x*)}]$	(Tan et al., 2002)
53.	Added value (AV)	$\max[p(y x) - p(*y), p(x y) - p(x*)]$	(Tan et al., 2002)
54.	Collective strength	$\frac{p(xy)+p(\bar{x}\bar{y})}{p(x*)p(y)+p(\bar{x}*)p(*y)} \cdot \frac{1-p(x*)p(*y)-p(\bar{x}*)p(*y)}{1-p(xy)-p(\bar{x}\bar{y})}$	(Tan et al., 2002)
55.	Klogsen	$\sqrt{p(xy)} \cdot AV$	(Tan et al., 2002)

Pecina P. A machine learning approach to multiword expression extraction[C] //Proceedings of the LREC Workshop Towards a Shared Task for Multiword Expressions (MWE 2008). Marrakech,[s. p.], 2008, 2008: 54-61.

降维

Dimensionality Reduction

- 维基百科: ~2千9百万英文文档, 词汇量: ~1百万词
 - 词-文档矩阵的高维性
 - 稀疏性/Sparsity
 - 行和列的次序无关
- 目标
 - 好的相似度衡量词或文本
 - 稠密表示/dense representation
- 稀疏向量和稠密向量
 - 短向量可能更容易用作机器学习中的特征(需要调整的权重更少)
 - 密集向量可能比存储显式计数更好地泛化
 - 可能在捕捉同义词synonymy方面做得更好
 - 实际效果更好



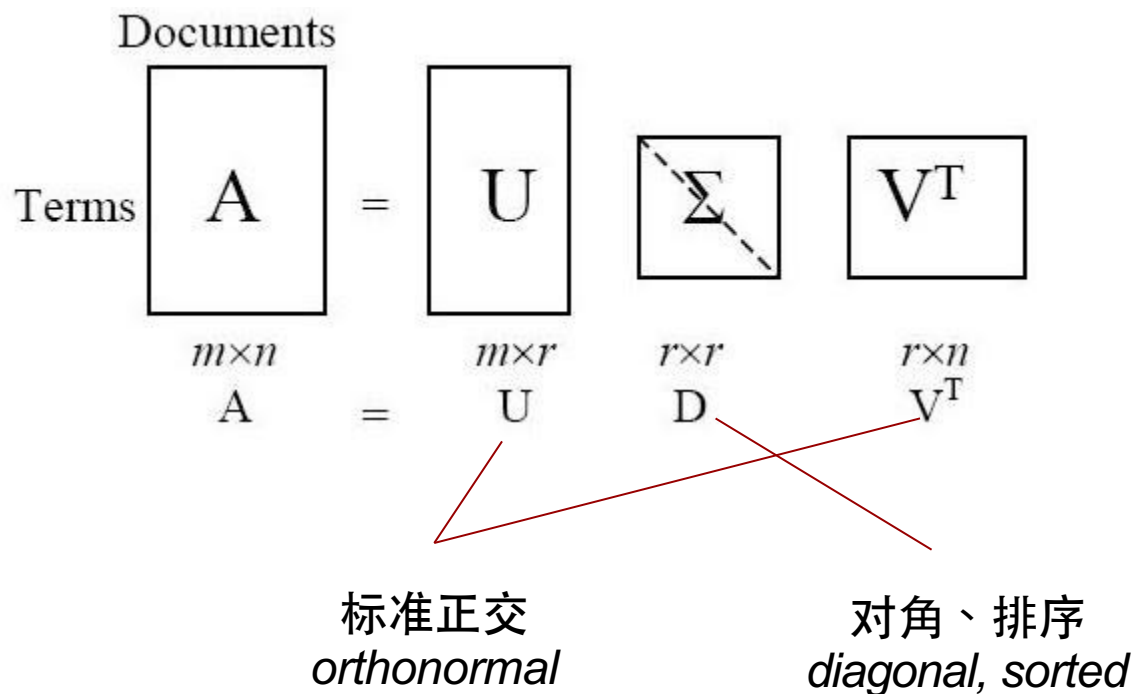
(Pecina'09)

A	0
a	0
aa	0
aal	0
aalii	0
aam	0
Aani	0
aardvark	1
aardwolf	0
...	0
zymotoxic	0
zymurgy	0
Zyrenian	0
Zyrian	0
Zyryan	0
zythem	0
Zythia	0
zythum	0
Zyzomys	0
Zyzzogeton	46 0

奇异值分解

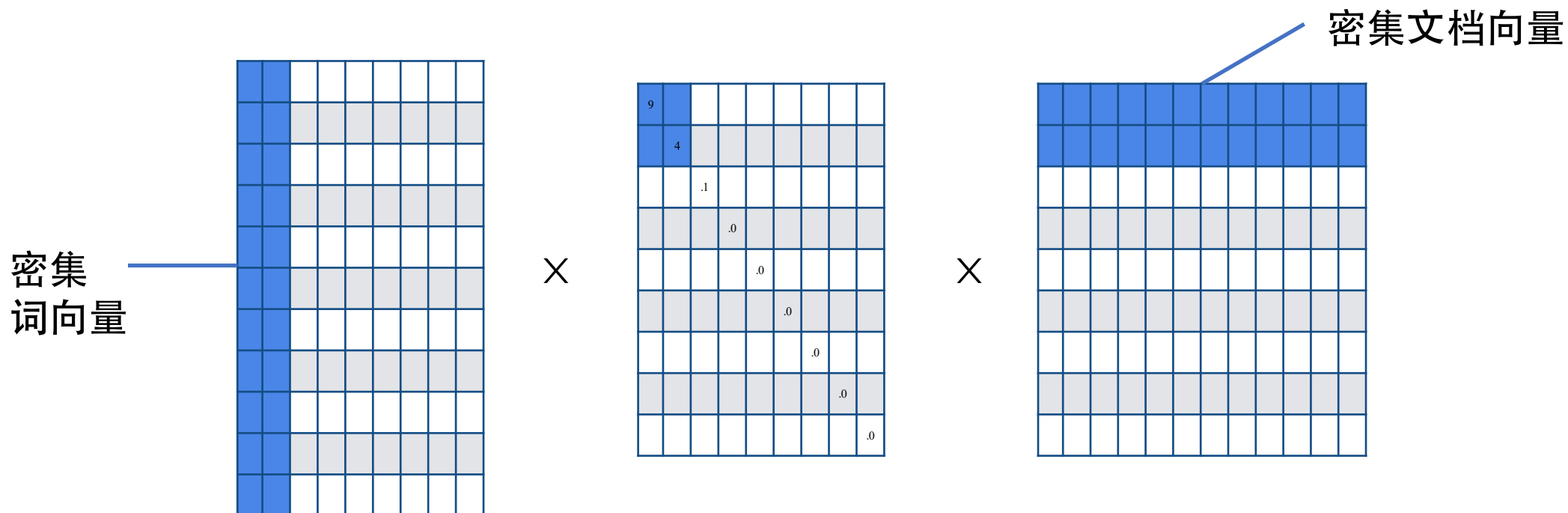
Singular Value Decomposition (SVD)

- 解决方案的思路
 - 寻找一个到低维空间的投影 (~300维)
 - 使得特征易于分离



截断的奇异值分解 Truncated SVD

- 只考虑对角矩阵中最左边的k项(k个最大的奇异值) 近似整个矩阵



$$A_{m \times n} \approx U_{m \times k} \Sigma_{k \times k} V_{k \times n}^T, \text{ 其中 } k \ll m, n$$

潜在语义分析

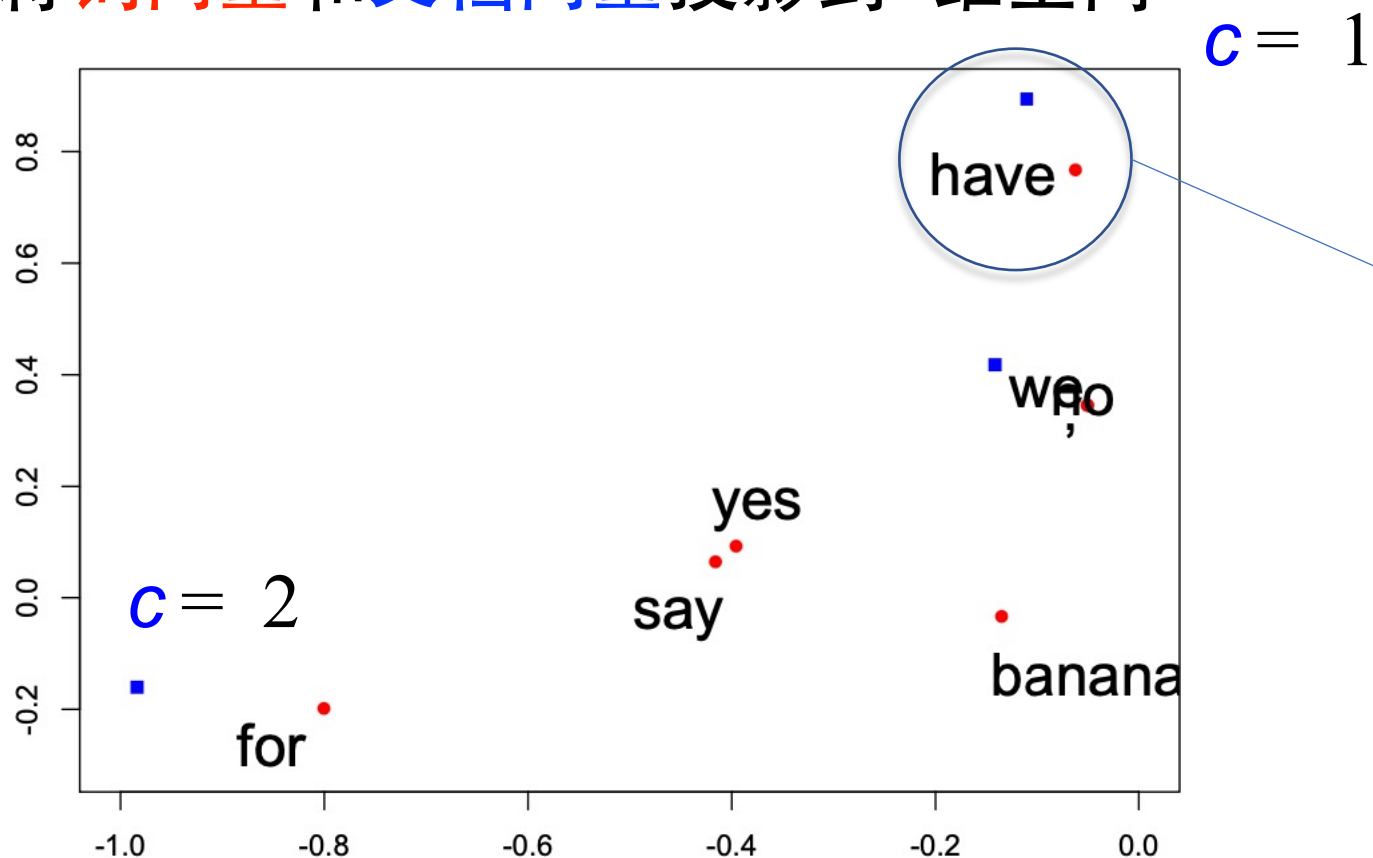
Latent Semantic Analysis (LSA)

#0	#1	#2	#3	#4	#5
we	music	company	how	program	10
said	film	mr	what	project	30
have	theater	its	about	russian	11
they	mr	inc	their	space	12
not	this	stock	or	russia	15
but	who	companies	this	center	13
be	movie	sales	are	programs	14
do	which	shares	history	clark	20
he	show	said	be	aircraft	sept
this	about	business	social	ballet	16
there	dance	share	these	its	25
you	its	chief	other	projects	17
are	disney	executive	research	orchestra	18
what	play	president	writes	development	19
if	production	group	language	work	21

Deerwester S, Dumais S T, Furnas G W, et al. Indexing by latent semantic analysis[J]. Journal of the American society for information science, 1990, 41(6): 391-407.

LSA例子

- 将词向量和文档向量投影到2维空间



	1	2	3
,	1	0	1
bananas	1	1	1
for	0	1	0
have	1	0	0
no	1	0	1
say	0	1	1
we	1	0	1
yes	1	1	0

A: count matrix

大纲

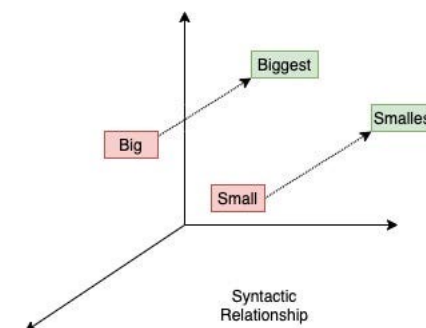
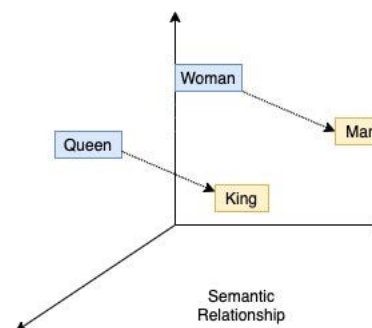
- 词汇语义学
- 电子词典
- 词的离散表示
- 词向量
- 词表示的评估

分布式表征

Distributed Representations

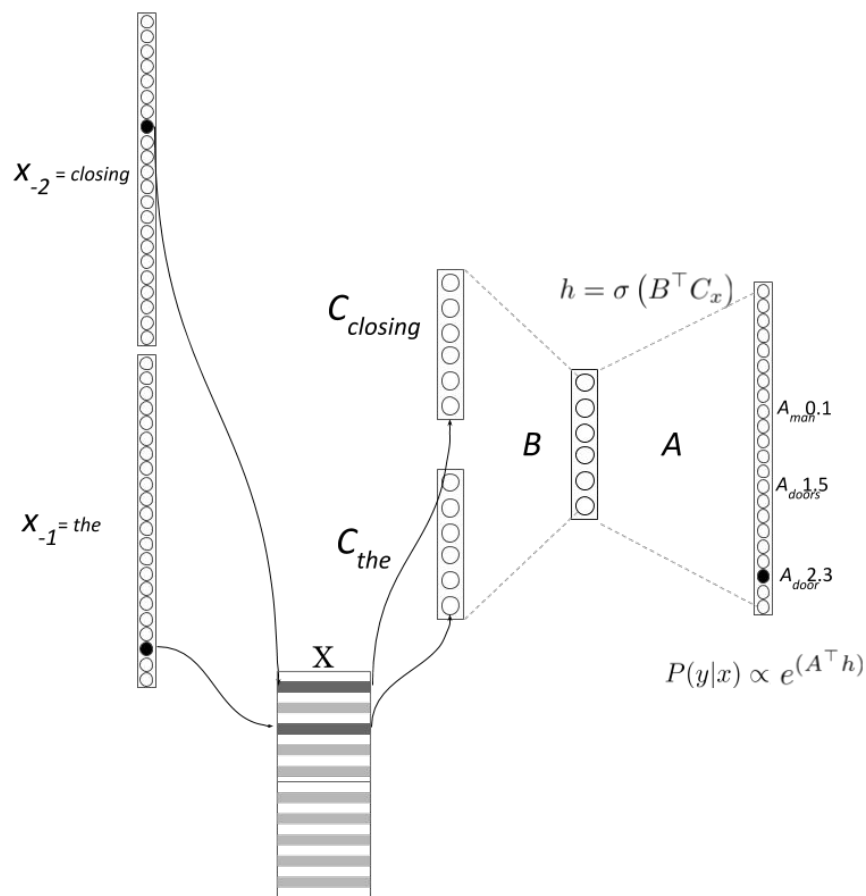
- 词向量/Word Vector

WORD	d1	d2	d3	d4	d5	...	d50
summer	0.12	0.21	0.07	0.25	0.33	...	0.51
spring	0.19	0.57	0.99	0.30	0.02	...	0.73
fall	0.53	0.77	0.43	0.20	0.29	...	0.85
light	0.00	0.68	0.84	0.45	0.11	...	0.03
clear	0.27	0.50	0.21	0.56	0.25	...	0.32
blizzard	0.15	0.05	0.64	0.17	0.99	...	0.23



<https://towardsdatascience.com/word2vec-research-paper-explained-205cb7eccc30/>

“独热/One-hot” 向量 vs. 词向量/嵌入 Word Vectors (Embeddings)

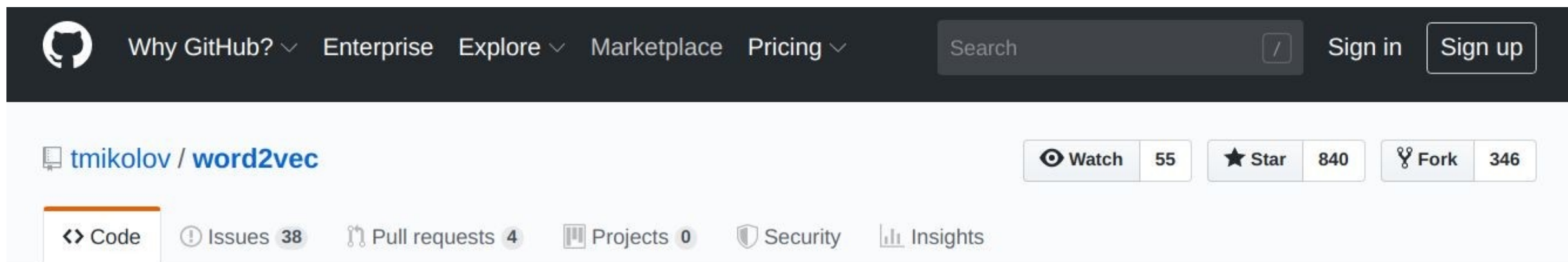


低维词表征

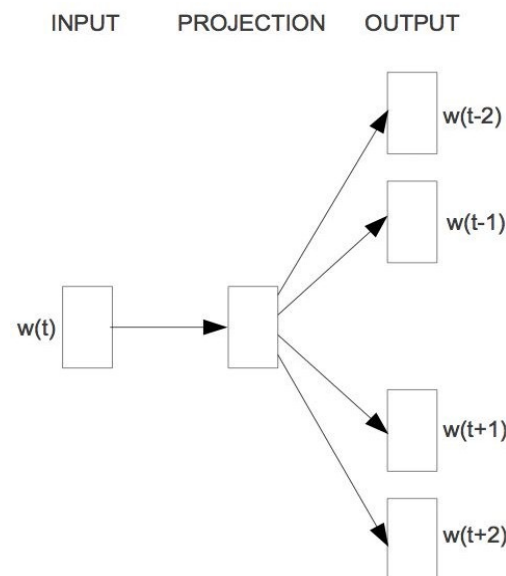
- Learning representations by back-propagating errors
 - Rumelhart, Hinton & Williams, 1986
- A neural probabilistic language model
 - Bengio et al., 2003
- Natural Language Processing (almost) from scratch
 - Collobert & Weston, 2008
- Word representations: A simple and general method for semi-supervised learning
 - Turian et al., 2010
- Distributed Representations of Words and Phrases and their Compositionality
 - Word2Vec; Mikolov et al., 2013

Word2Vec

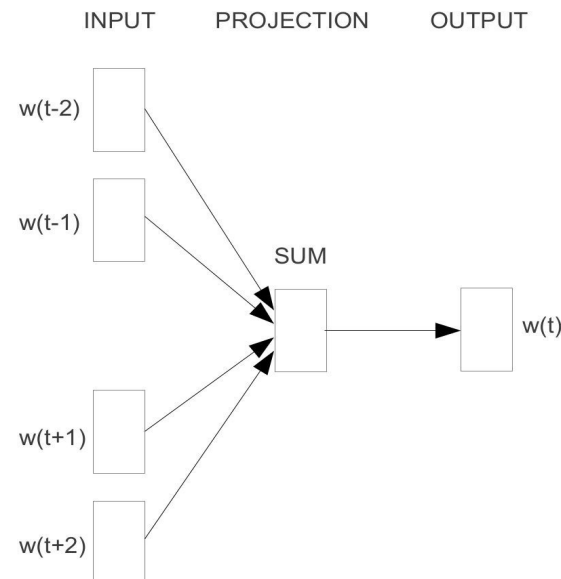
- 流行的词嵌入方法
- 训练速度非常快
- 代码可在网上获得
- 思路：预测而不是计数



Word2Vec



Skip-gram
跳元语法



CBOW (Continuous Bag-of-Words Model)
连续词袋模型

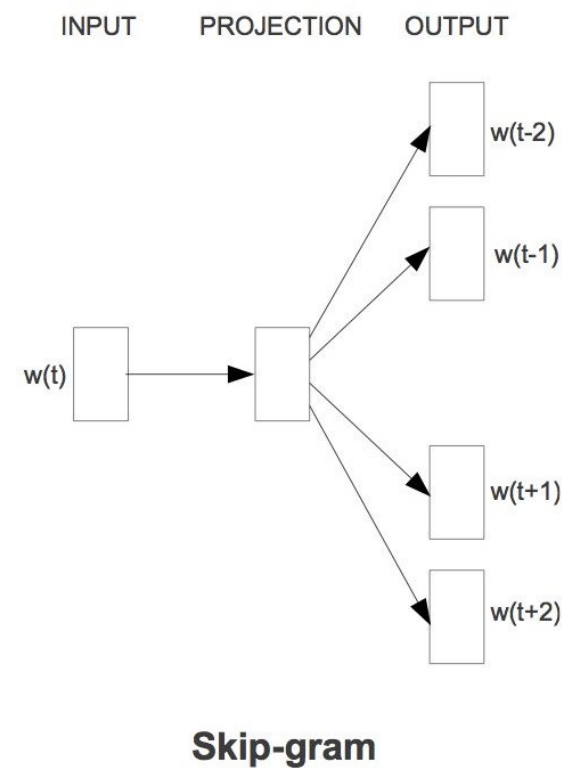
Tomás Mikolov, Ilya Sutskever, Kai Chen, Gregory S. Corrado, Jeffrey Dean: Distributed Representations of Words and Phrases and their Compositionality. NIPS 2013: 3111-3119

2023 NeurIPS Test of Time Award

Skip-gram Prediction

- Predict vs. Count

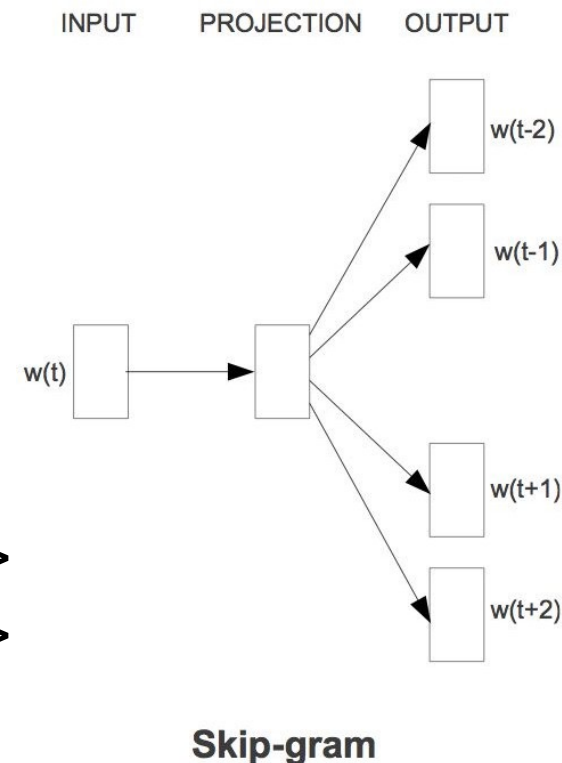
the cat sat on the mat



Skip-gram Prediction

- Predict vs. Count

the cat sat on the mat

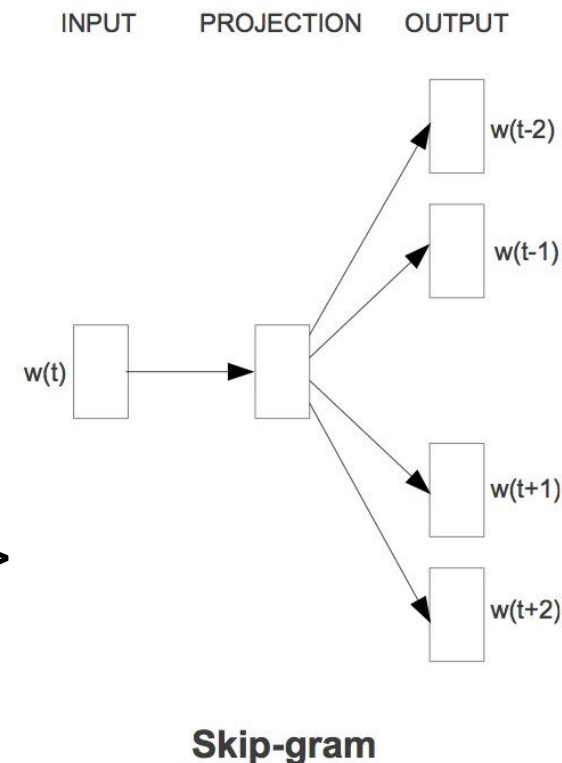


context size = 2

Skip-gram Prediction

- Predict vs. Count

the cat sat on the mat

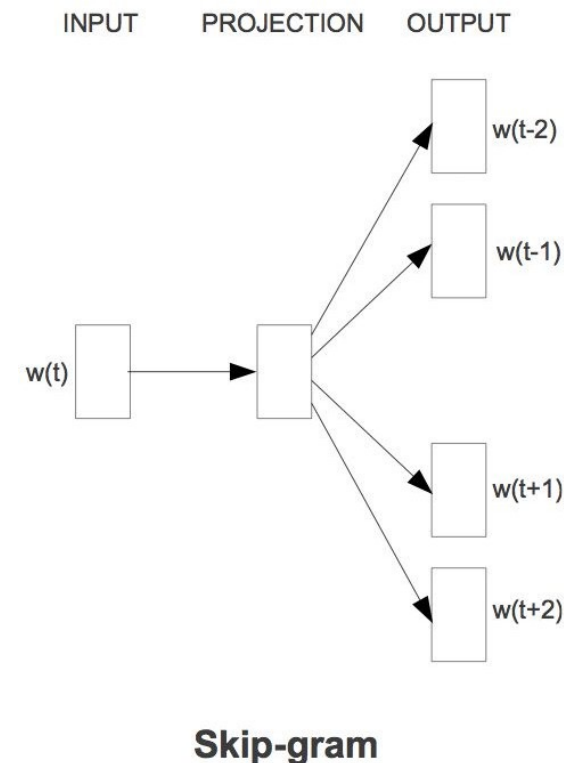


context size = 2

Skip-gram Prediction

- Predict vs. Count

the cat sat on the mat

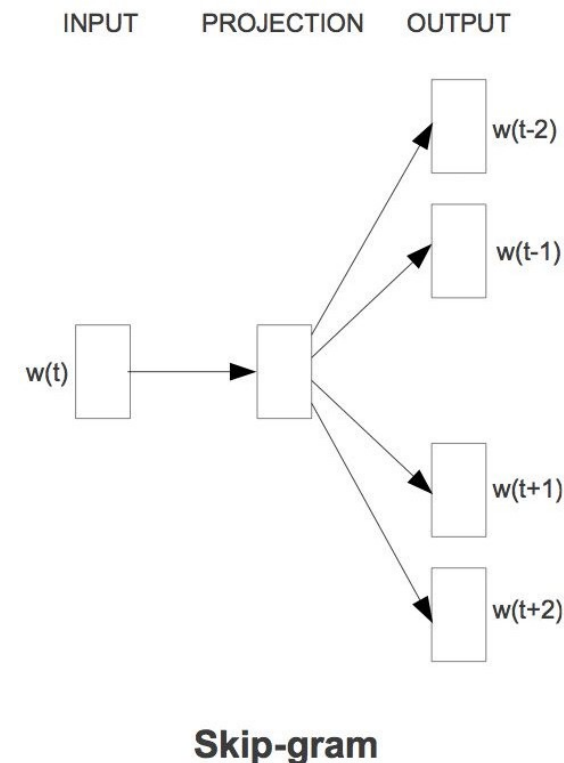


context size = 2

Skip-gram Prediction

- Predict vs. Count

the cat sat on the mat

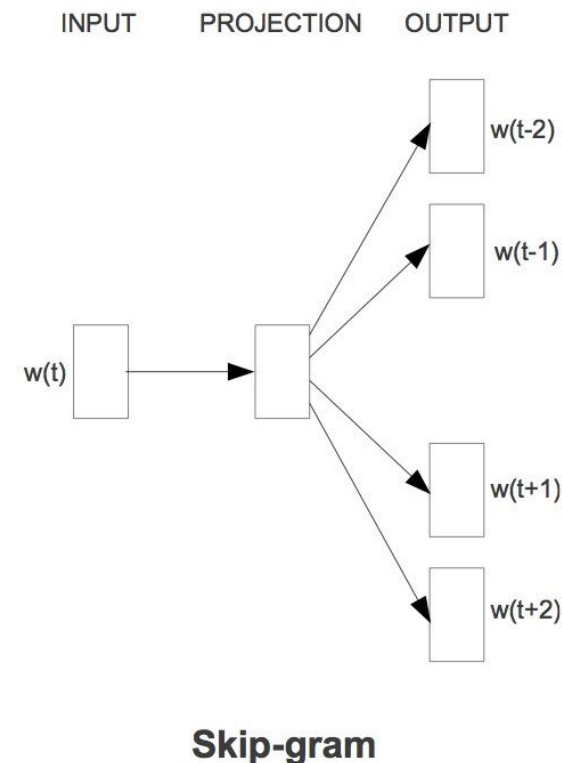


context size = 2

Skip-gram Prediction

- Predict vs. Count

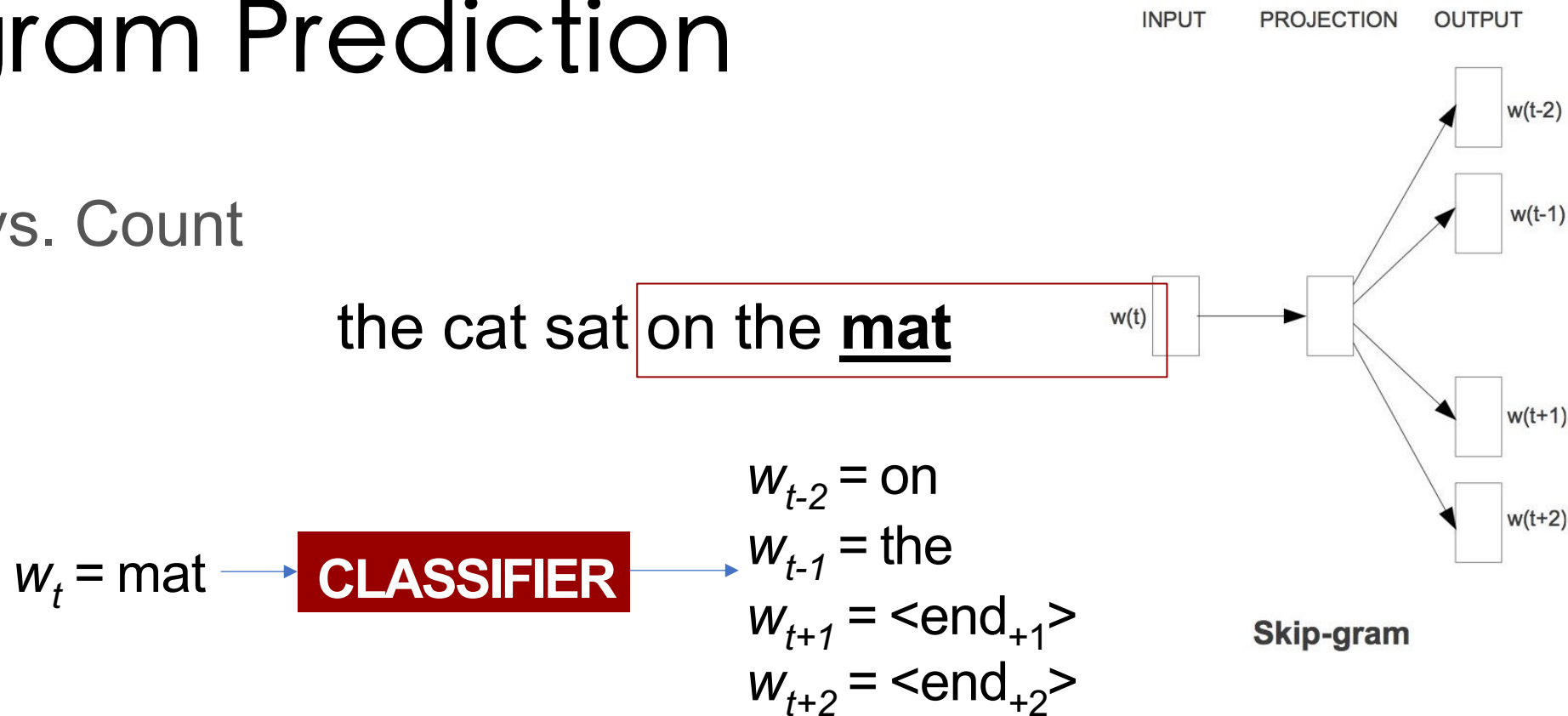
the cat sat on the mat



context size = 2

Skip-gram Prediction

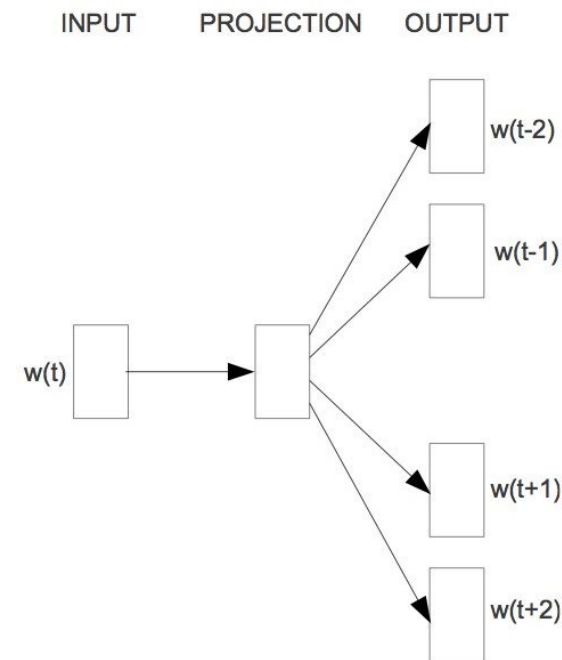
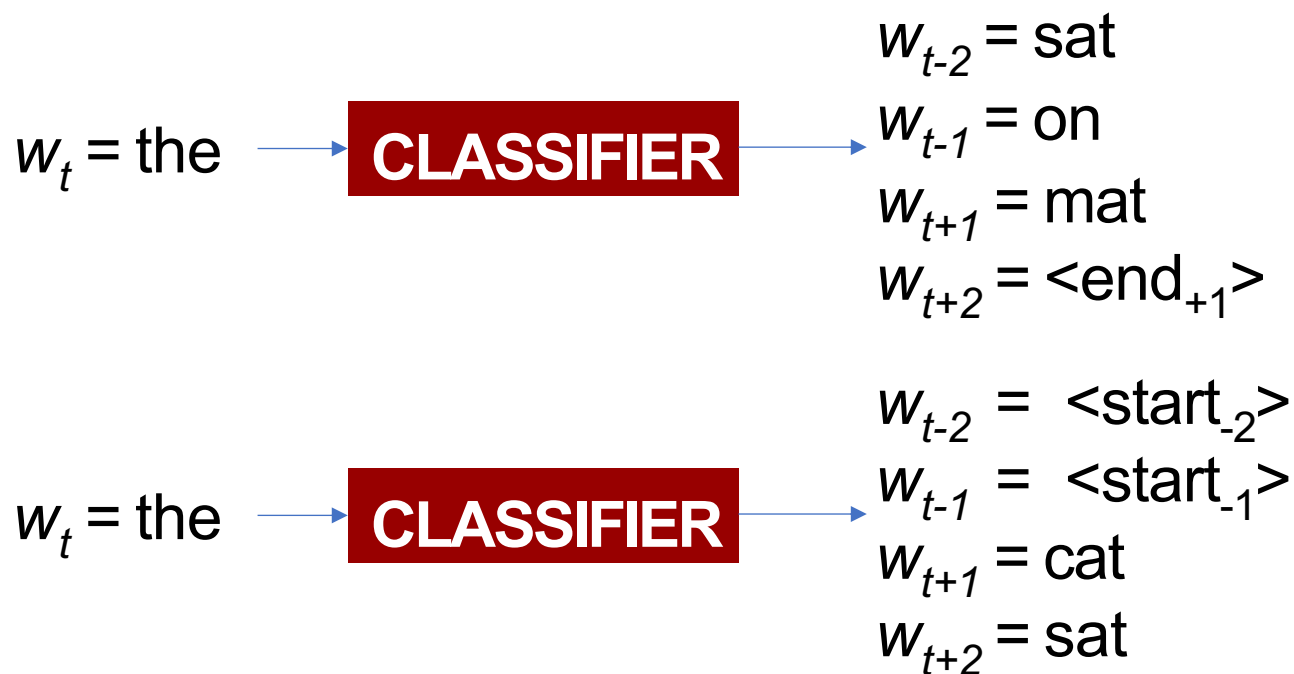
- Predict vs. Count



context size = 2

Skip-gram Prediction

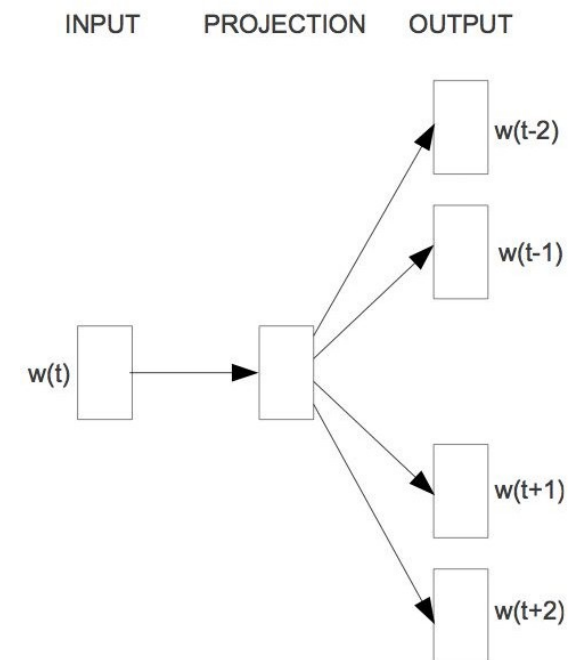
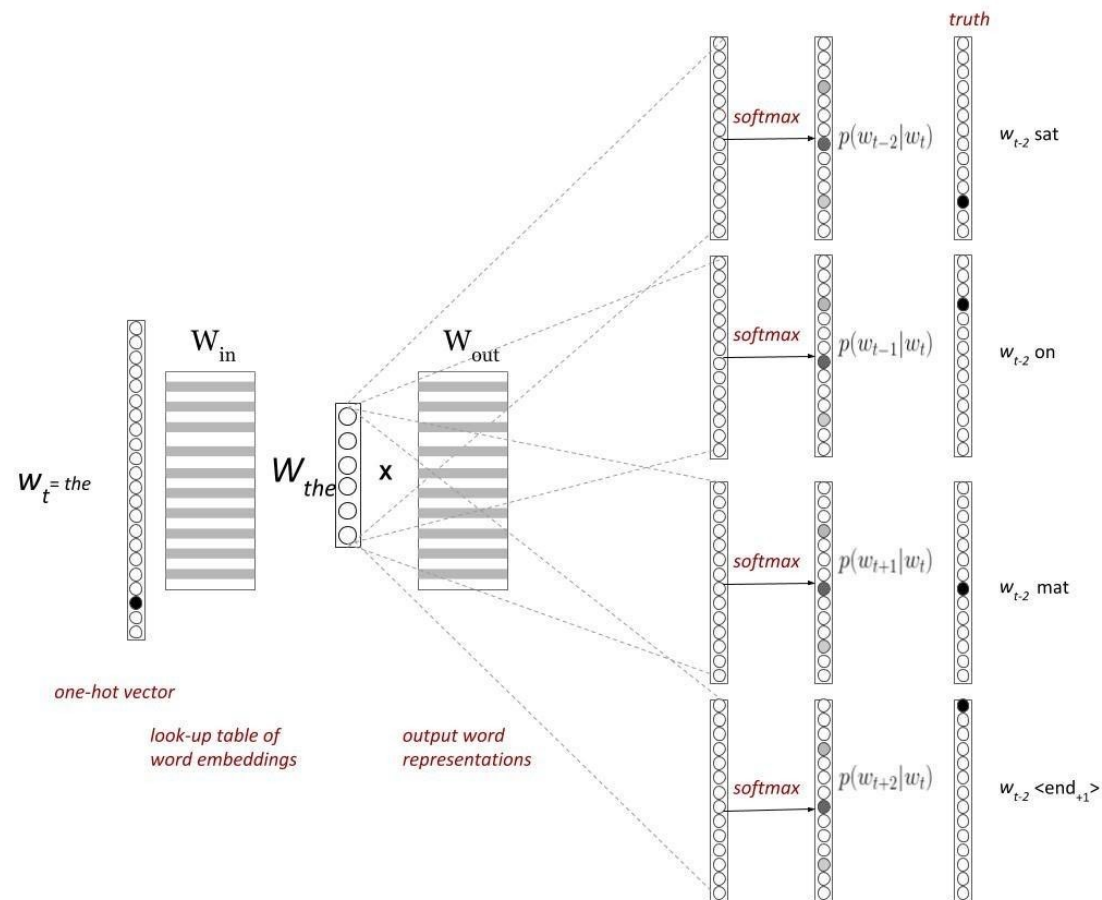
- 同一个词可以出现在不同的上下文



Skip-gram

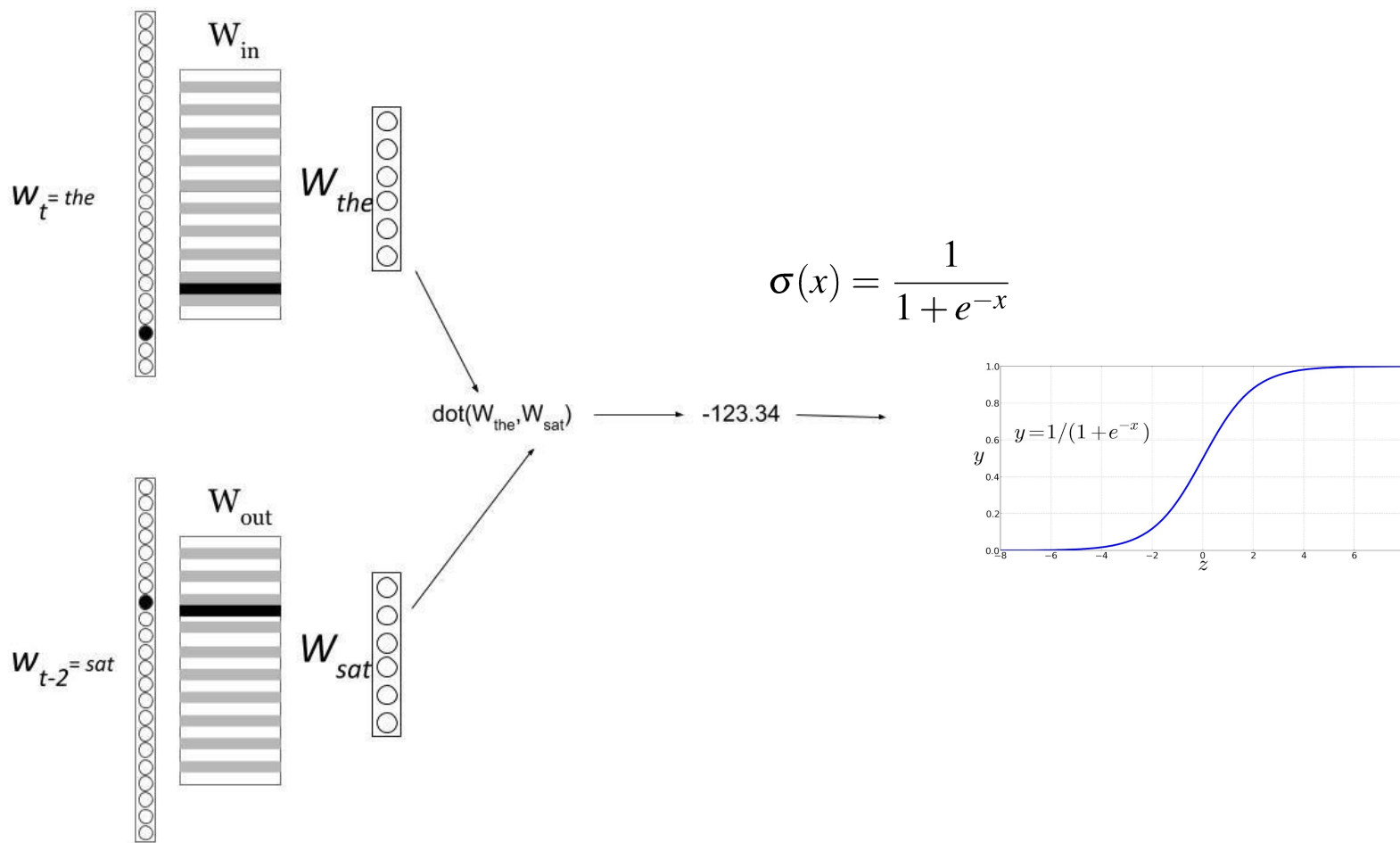
context size = 2

Skip-gram Prediction



Skip-gram

如何计算 $p(+|t, c)$



Skip-gram Objective

- 对于语料中的每个词, 最大化给定当前中心单词的任何上下文窗口的概率

$$J(\Theta) = \prod_{t=1}^T \prod_{-m \leq j \leq m, j \neq 0} p(w_{t+j} | w_t; \Theta)$$

$$J(\Theta) = -\frac{1}{T} \sum_{t=1}^T \sum_{-m \leq j \leq m, j \neq 0} \log p(w_{t+j} | w_t; \Theta)$$

- 其中 $p(w_{t+j} | w_t) = p(o|c) = \frac{\exp(u_o^\top u_c)}{\sum_{i=1}^V \exp(u_i^\top u_c)}$
 - o : 外围(上下文)词下标
 - c : 中心词下标(w_t)
 - V : 词表大小, $50K \sim 30M$

Skip-gram: 负采样

- 问题1: $V = 50K \sim 30M$, 太大!

$$p(w_{t+j}|w_t) = p(o|c) = \frac{\exp(u_o^\top u_c)}{\sum_{i=1}^V \exp(u_i^\top u_c)}$$

- 方案: 负采样 (Negative sampling)
 - 将中心词和邻近的上下文词形成正样本对
 - 中心词与词典中其他词, 形成负样本对

(banking, regulation)

(banking, ongchoi)

Skip-gram: 负采样

- 将多分类问题转化成二分类问题

$$p(o|c) = \frac{\exp(u_o^\top u_c)}{\sum_{i=1}^V \exp(u_i^\top u_c)} \rightarrow p(o|c) = \frac{1}{1 + \exp(-u_o^\top u_c)} = \sigma(u_o^\top u_c)$$

- 新目标 (一个上下文词， k 个负样本):

$$\log p(o_+ | c) + \sum_{i=1}^k \log(1 - p(o_i | c))$$

Skip-gram: 如何选择负样本

- 根据一元模型的词率($p(w)$)选取负样本
- 通常的选择方案:

$$p_{\alpha}(w) = \frac{\text{count}(w)^{\alpha}}{\sum_w \text{count}(w)^{\alpha}}$$

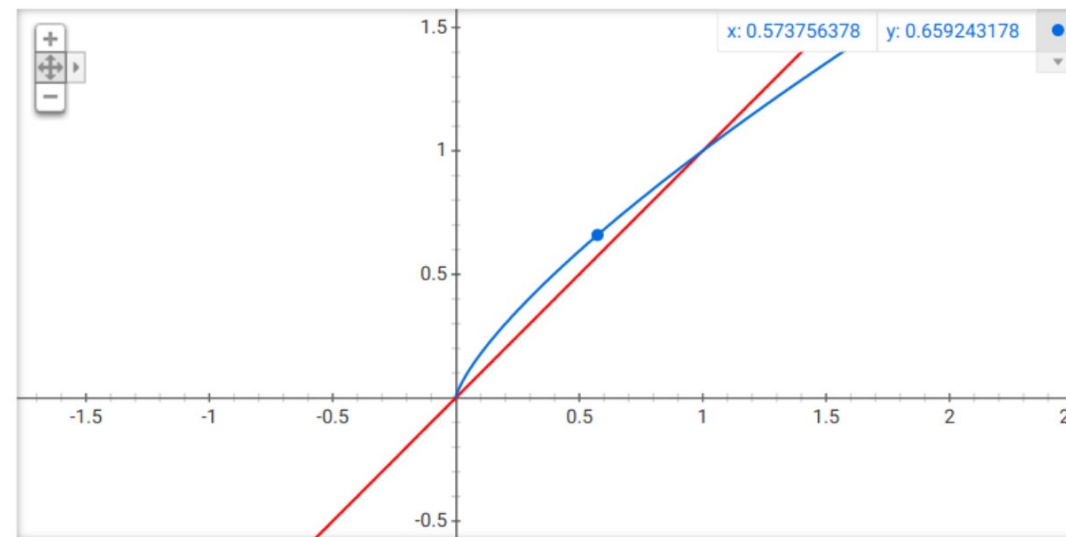
- 经验值: 取 $\alpha = 0.75$ 效果更佳
- 作用: 给低频词稍微高的概率

- e.g., $p(a) = 0.99, p(b) = 0.01$

$$p_{\alpha}(a) = \frac{0.99^{0.75}}{0.99^{0.75} + 0.01^{0.75}} = 0.97$$

$$p_{\alpha}(b) = \frac{0.01^{0.75}}{0.99^{0.75} + 0.01^{0.75}} = 0.03$$

Graph for $x^{(3/4)}$, x



In-Class Practice III

可用的稠密词嵌入

- Word2vec (Mikolov et al. 2013)
 - <https://code.google.com/archive/p/word2vec/>
- GloVe (Pennington et al. 2014)
 - <http://nlp.stanford.edu/projects/glove/>
- Fasttext (Bojanowski et al. 2017)
 - <http://www.fasttext.cc/>

GloVe: Global Vectors for Word Representation

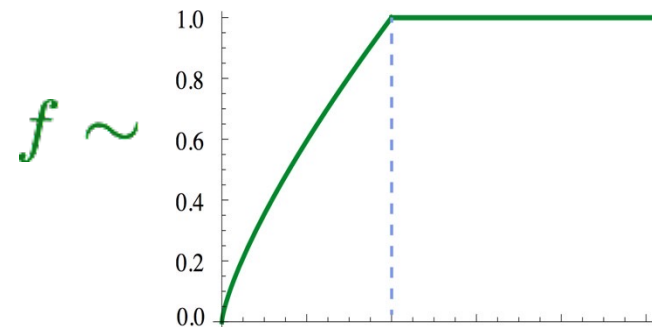
全局词向量模型

- 核心创新：融合全局 + 局部
 - 核心假设: 词向量相似度 $\approx$ 词共现概率的对数关系

$$J = \sum_{i,j=1}^V f(X_{ij})(w_i^\top \tilde{w}_j + b_i + \tilde{b}_j - \log X_{ij})^2$$

其中

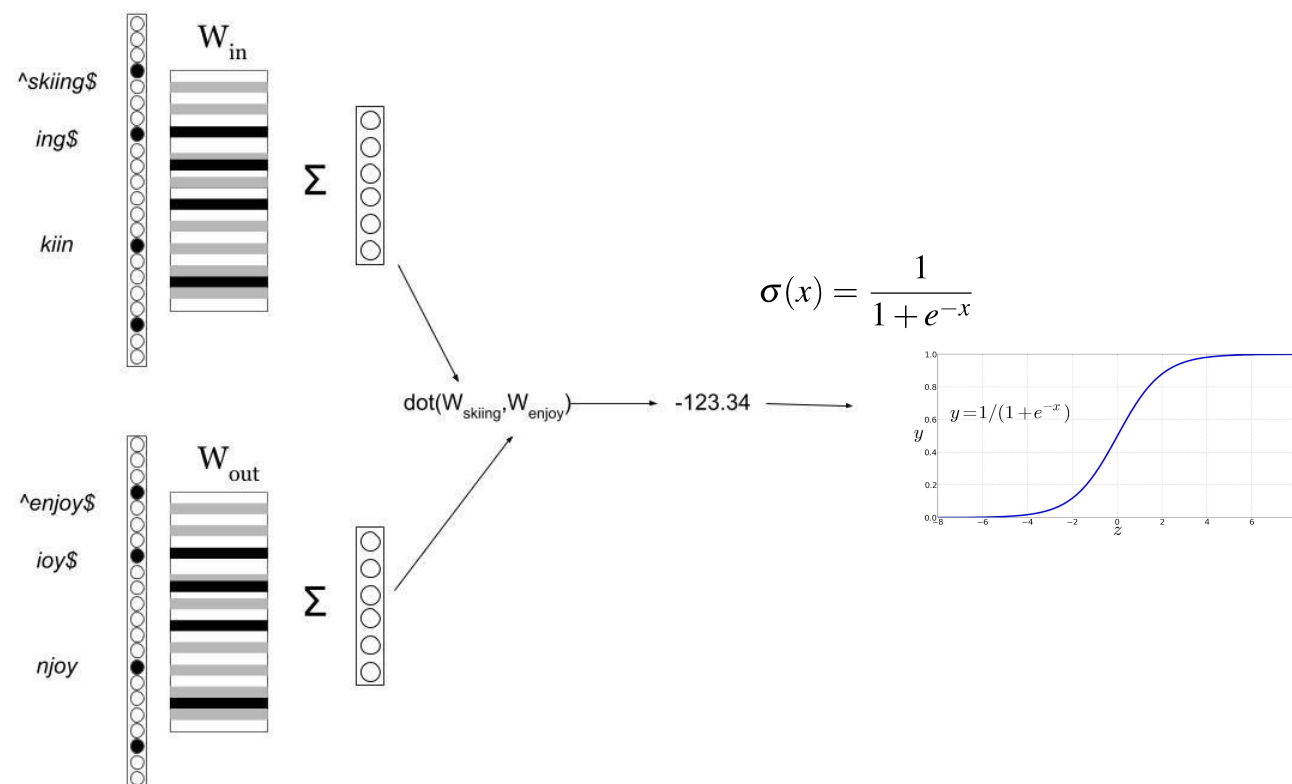
- X_{ij} : 词贡献概率
- $w_i/\tilde{w}_j$: 词/(上下文)词向量
- $b_i/\tilde{b}_j$: 偏移项bias



Jeffrey Pennington, Richard Socher, Christopher D. Manning:
GloVe: Global Vectors for Word Representation. EMNLP 2014: 1532-1543
2024 ACL Test of Time Award

FastText

- 引入子词(Subword)机制



Piotr Bojanowski, Edouard Grave, Armand Joulin, Tomáš Mikolov: Enriching Word Vectors with Subword Information. Trans. Assoc. Comput. Linguistics 5: 135-146 (2017)

Word2Vec, GloVe, FastText对比

	Word2Vec(Skip-gram/CBOW)	GloVe (全局词向量)	FastText
提出时间 & 机构	2013 年，Google (Mikolov 等人)	2014 年，斯坦福大学 (C. Manning 等人)	2016 年，Facebook AI Research (Mikolov等人)
核心依赖	仅局部上下文窗口(无全局统计)	全局词共现统计 + 局部上下文	字符级 n-gram + Word2Vec 框架
核心优势	结构简洁、训练速度快、扩展性好	语义精准、数学解释性强、全局信息融合	1. 天然支持 OOV(未登录词); 2. 对稀有词、拼写错误词鲁棒; 3. 训练效率高
关键局限	1. 缺乏全局统计信息; 2. OOV 处理弱	1. 训练依赖共现矩阵(空间开销大); 2. OOV 处理弱	1. 词向量维度较高(子词冗余); 2. 语义精准度略逊于 GloVe
数学解释性	较弱	强	中等
核心荣誉	NeurIPS 2023 十年时间检验奖	ACL 2024 十年时间检验奖	无顶级会议时间检验奖，但工业界应用广泛

大纲

- 词汇语义学
- 电子词典
- 词的离散表示
- 词向量
- 词表示的评估

词向量的评估

- 外部/Extrinsic
- 内部/Intrinsic
- 定性/Qualitative

WORD	d1	d2	d3	d4	d5	...	d50
summer	0.12	0.21	0.07	0.25	0.33	...	0.51
spring	0.19	0.57	0.99	0.30	0.02	...	0.73
fall	0.53	0.77	0.43	0.20	0.29	...	0.85
light	0.00	0.68	0.84	0.45	0.11	...	0.03
clear	0.27	0.50	0.21	0.56	0.25	...	0.32
blizzard	0.15	0.05	0.64	0.17	0.99	...	0.23

外部评估

Extrinsic Evaluation

- 外部评价/Extrinsic evaluation：测试表征是否对下游任务有效
 - 提供嵌入作为同一分类器的输入，带有预训练嵌入的模型表现如何？
- 常见的下游任务：
 - 组块分析/Chunking
 - 词性标注/POS tagging
 - 句法分析/Parsing
 - 机器翻译/MT
 - 语义角色标注/SRL
 - 主题分类/Topic categorization
 - 情感分析/Sentiment analysis
 - 隐喻检测/Metaphor detection
 - ...

内部评估

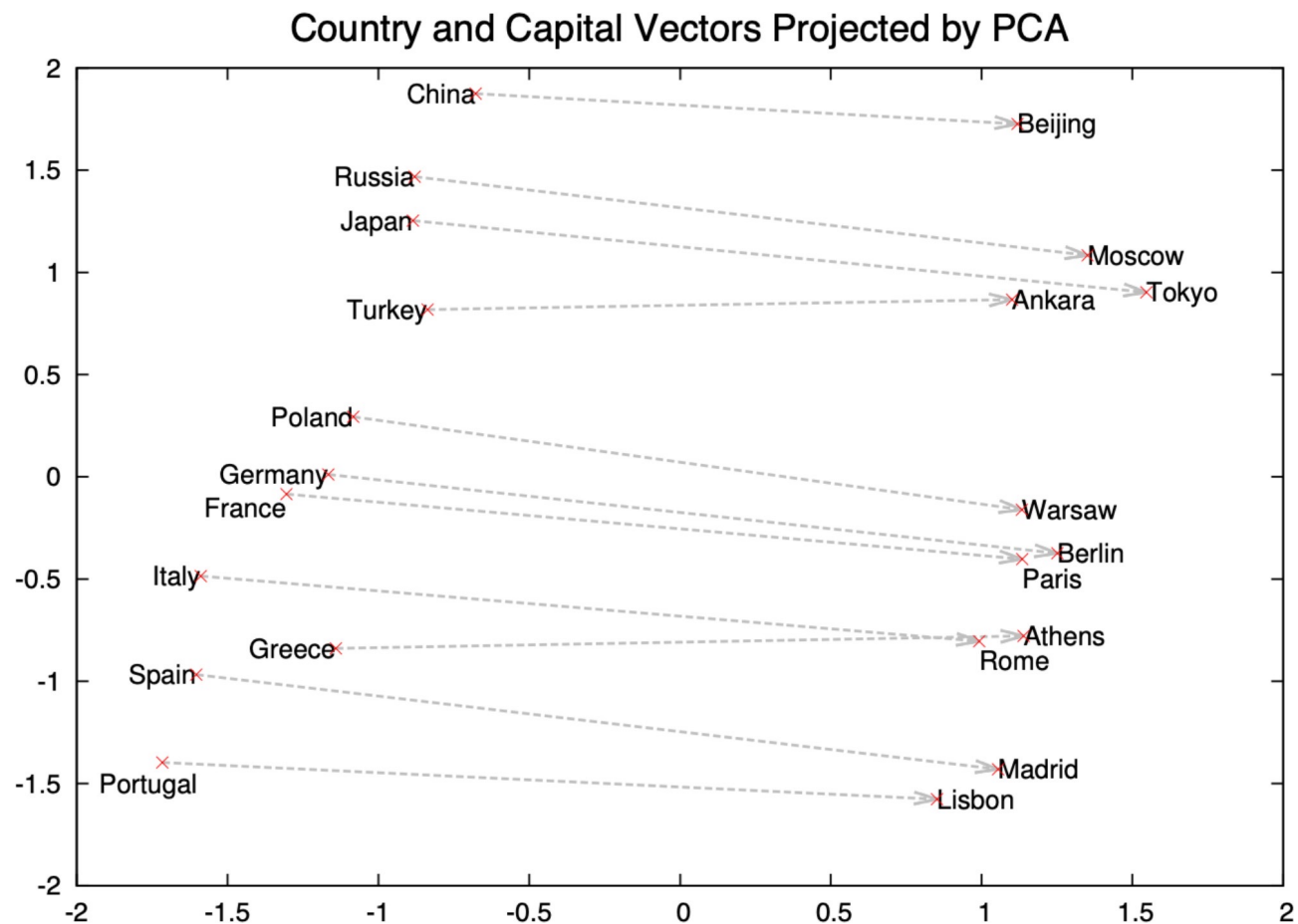
Intrinsic Evaluation

- 内部评价/*Intrinsic evaluation*: 测试表征是否与我们对词义的直觉一致
 - 词嵌入的余弦相似度与人类判断的相关性有多近似？
 - 类比完成: $a:b \leftrightarrow c:?$

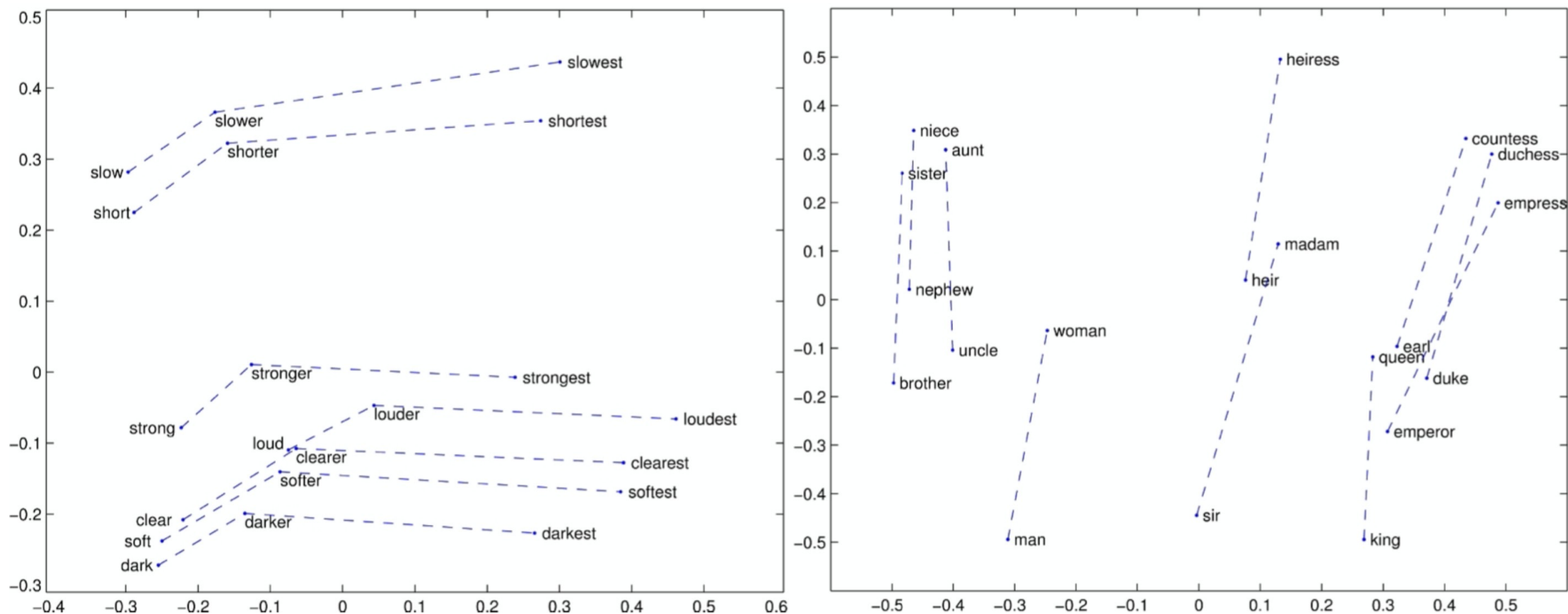
A:B<->C:?

man:woman :: king:?

$$d = \arg \max_i \frac{(x_b - x_a + x_c)^T x_i}{||x_b - x_a + x_c||}$$



A:B<->C:?



内部评估

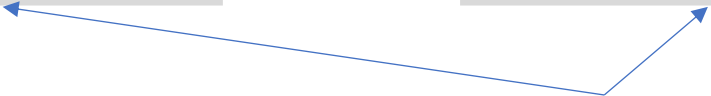
Intrinsic Evaluation

word1	word2	similarity (humans)
vanish	disappear	9.8
behave	obey	7.3
belief	impression	5.95
muscle	bone	3.65
modest	flexible	0.98
hole	agreement	0.3

similarity (embeddings)
1.1
0.5
0.3
1.7
0.98
0.3

- WS-353 (Finkelstein et al. 2002)
- MEN-3k (Bruni et al. 2012)
- SimLex-999 dataset (Hill et al., 2015)

Spearman's rho (human ranks, model ranks)



可视化

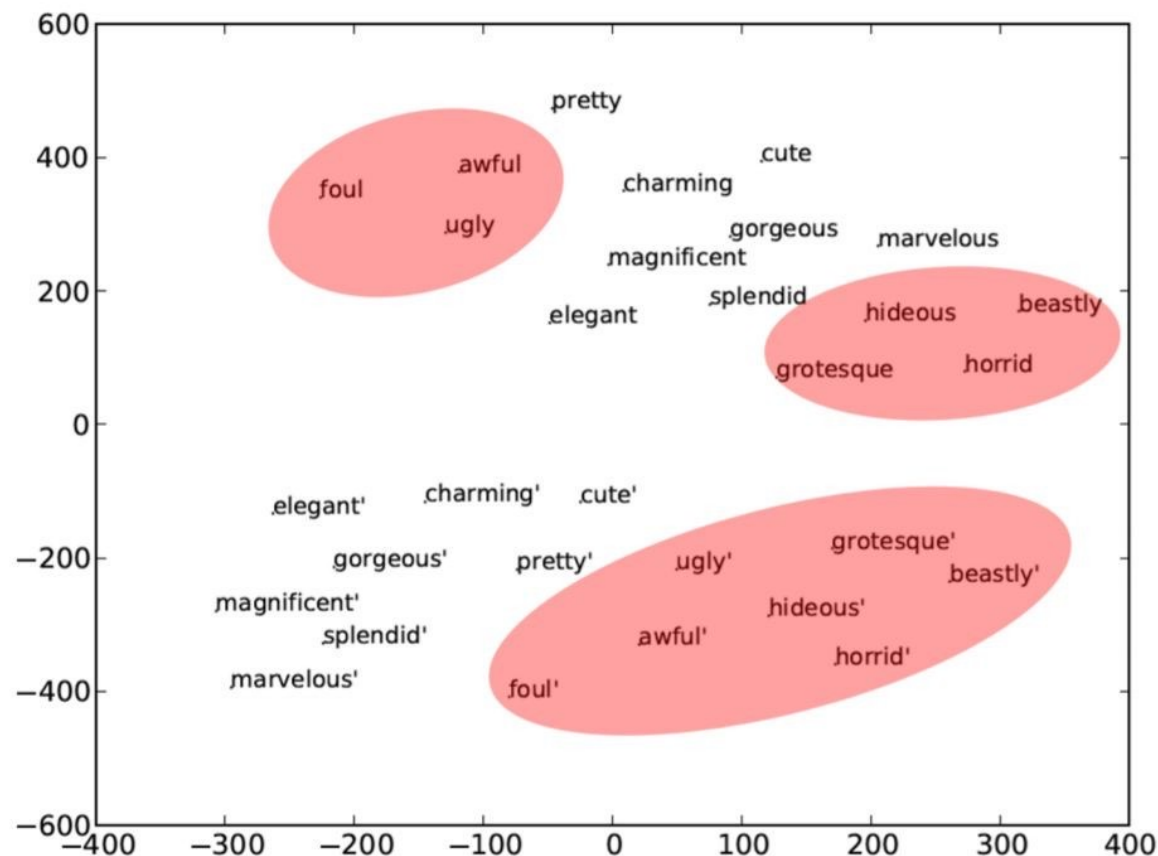
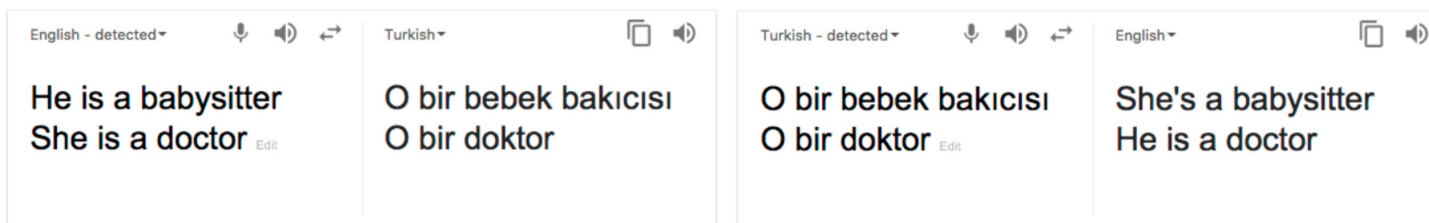


Figure 6.5: Monolingual (top) and multilingual (bottom; marked with apostrophe) word projections of the antonyms (shown in red) and synonyms of “beautiful”.

- L.J.P. van der Maaten and G.E. Hinton. Visualizing Data Using t-SNE. Journal of Machine Learning Research 9(Nov):2579-2605, 2008
- Manaal Faruqui, Jesse Dodge, Sujay Kumar Jauhar, Chris Dyer, Eduard H. Hovy, Noah A. Smith: Retrofitting Word Vectors to Semantic Lexicons. HLT-NAACL 2015: 1606-1615

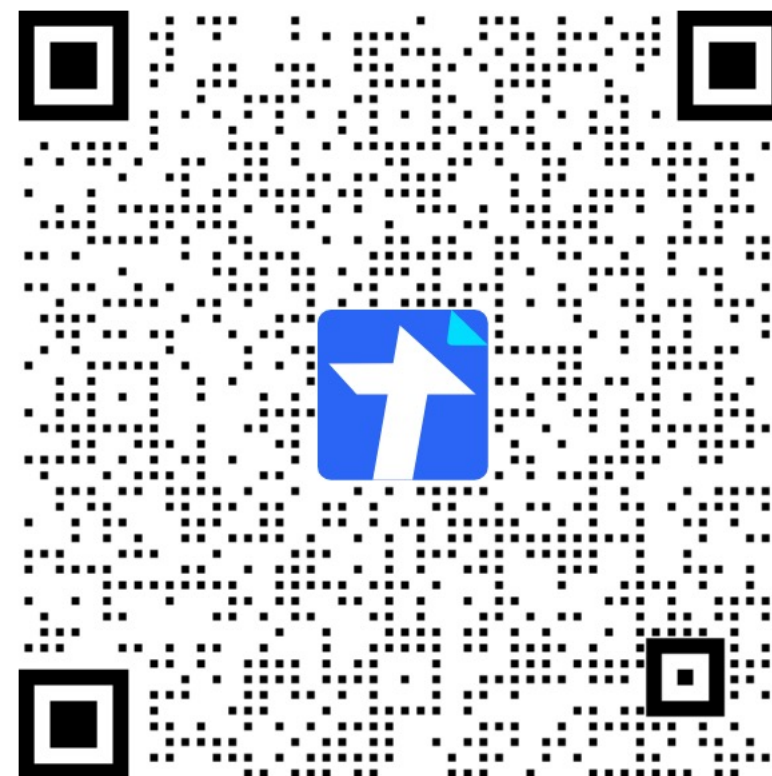
其他主题

- 词嵌入的偏见/bias (性别偏见)



- 多语言词嵌入/Multilingual word embeddings
- 预先训练的语境化词嵌入 (例如 , Elmo, bert, Roberta)

一句话总结



- 课外阅读
 - [SLP3] Ch. 6

附录

1. 信息熵、互信息(逐点互信息、正点互信息)、KL散度、交叉熵的关系和性质

信息熵

- 信息熵（entropy）指的是某条消息所含的信息量
- 信息熵的计算方法如下:
- 例子: 单次抛硬币试验结果的信息熵:

$$H(X) = - \sum_x P(x) \log P(x)$$

- $H(X) = P(x = \text{正}) \log P(x = \text{正}) + P(x = \text{反}) \log P(x = \text{反})$

- $= - \left(\frac{1}{2} \log \frac{1}{2} + \frac{1}{2} \log \frac{1}{2} \right)$

- $= - \log \frac{1}{2} = 1$

互信息、逐点互信息、正点互信息

- 互信息(MI: Mutual Information): 两个离散型随机变量 X 与 Y 相关程度的度量

$$\begin{aligned} \text{MI}(X, Y) &= \sum_{x, y} P(x, y) \log \frac{P(x, y)}{P(x)P(y)} \\ &= E_{P(x, y)} \log \frac{P(x, y)}{P(x)P(y)} \end{aligned}$$

- 逐点互信息(PMI: Pointwise Mutual Information): 两个事件/词的关联度

$$\text{PMI}(x, y) = \log \frac{P(x, y)}{P(x)P(y)}$$

- 正点互信息(PPMI: Pointwise Mutual Information): 聚焦有效关联

$$\text{PPMI}(x, y) = \max \left(\log_2 \frac{P(x, y)}{P(x)P(y)}, 0 \right)$$

KL散度与交叉熵

- **KL散度(Kullback-Leibler Divergence)**: 衡量两个分布 P 与 Q 的“差异”:

$$D(P\|Q) = \sum_x P(x) \log[P(x)/Q(x)]$$

- 性质
 - 非对称
 - 非负

- **交叉熵**: 用 Q 近似 P 的编码代价:

$$\begin{aligned} H(P, Q) &= - \sum_x P(x) \log Q(x) \\ &= H(P) + D(P\|Q) \end{aligned}$$

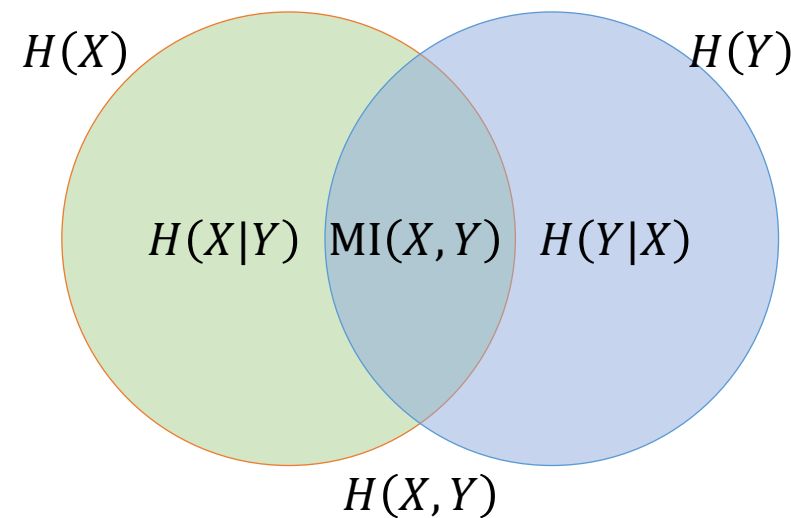
熵、互信息、逐点互信息的关系

- 互信息: X 与 Y 共享的信息量

$$\begin{aligned} \text{MI}(X, Y) &= H(X) - H(X|Y) \\ &= H(Y) - H(Y|X) \end{aligned}$$

- 互信息是逐点互信息的期望

$$\begin{aligned} \text{MI}(X, Y) &= \sum_{x,y} P(x, y) \log \frac{P(x, y)}{P(x)P(y)} \\ &= E_{P(x,y)} \log \frac{P(x,y)}{P(x)P(y)} = E[\text{PMI}(x, y)] \end{aligned}$$



关系：熵、KL散度与交叉熵；互信息、KL散度

$$H(P, Q) = H(P) + D(P \| Q)$$

- 交叉熵是 “真实分布 P 的熵” + “ P 与 Q 的差异（KL 散度）”
- 当 $P = Q$ 时， $D(P \| Q) = 0$ ， $H(P, Q) = H(P)$ （交叉熵 = 熵）
- 最小化交叉熵等价于最小化 KL 散度（因 $H(P)$ 为常数）

$$MI(X, Y) = D(P(X, Y) \| P(X)P(Y))$$

- 互信息是联合分布与边缘分布乘积的 KL 散度
- 衡量“ X 与 Y 独立” ($P(X, Y) = P(X)P(Y)$)的偏离程度

概念及关键性质总览

概念	关键性质	
信息熵	1. 非负 2. 均匀分布时熵最大	3. 熵的链式法则: $H(X, Y) = H(X) + H(Y X) = H(Y) + H(X Y)$
互信息	1. 非负 2. 对称: $MI(X, Y) = MI(Y, X)$	3. $MI(X, Y) \leq \min(H(X), H(Y))$ 4. 独立时为0
KL散度	1. 非负(仅 $P = Q$ 时为0) 2. 非对称: $D(P Q) \neq D(Q P)$	3. 不满足三角不等式
交叉熵	1. 非负	2. 当 $Q(x) = 0, P(x) > 0$ 时, $H(P, Q) = \infty$
PPMI/PMI	1. PMI可正可负为零(正/负/无相关)	2. PPMI非负 3. 独立时, $PMI=PPMI=0$

In-class Practice I: 余弦相似度

Cosine Similarity

- 已知

	pie	data	computer
cherry	442	8	2
digital	5	1683	1670
information	5	3982	3325

- 求词cherry与information, digital与information的余弦相似度

Ans. to In-class Practice I: 余弦相似度

Cosine Similarity

解: 已知 $\cos(\mathbf{v}, \mathbf{w}) = \frac{\mathbf{v} \cdot \mathbf{w}}{\|\mathbf{v}\| \|\mathbf{w}\|} = \frac{\sum_{i=1}^d v_i w_i}{\sqrt{\sum_{i=1}^d v_i^2} \sqrt{\sum_{i=1}^d w_i^2}}$

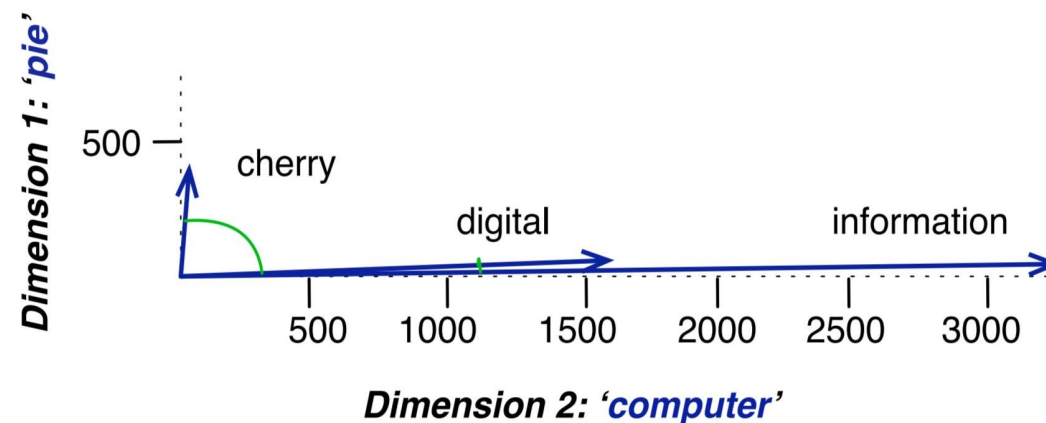
$\cos(\text{cherry}, \text{information})$

$$= \frac{442 * 5 + 8 * 3982 + 2 * 3325}{\sqrt{442^2 + 8^2 + 2^2} \sqrt{5^2 + 3982^2 + 3325^2}}$$
$$= 0.018$$

$\cos(\text{digital}, \text{information})$

$$= \frac{5 * 5 + 1683 * 3982 + 1670 * 3325}{\sqrt{5^2 + 1683^2 + 1670^2} \sqrt{5^2 + 3982^2 + 3325^2}}$$
$$= 0.996$$

	pie	data	computer
cherry	442	8	2
digital	5	1683	1670
information	5	3982	3325



In-class Practice II: PPMI

- 已知某语料库包含两类文本：体育类(30篇)、科技类(20篇)，共50篇。
核心词“进球”和“算法”的出现统计如下：

统计项	体育类	科技类	总计
包含“进球”文本数	25	1	26
不包含“进球”文本数	5	19	24
包含“算法”文本数	2	18	20
不包含“算法”文本数	28	2	30

- 计算词“进球”与文本类别的互信息(以2为底的对数；保留2位小数)
- 简要判断：“进球”和“算法”哪个词与文本类别更相关？（可逻辑推理）

Ans. to In-class Practice II: PPMI

解: 定义随机变量:

- X (文本类别, $X = 1$: 体育类, $X = 2$: 科技类),
- Y (词的出现状态, $Y = 1$: 包含该词, $Y = 0$: 不包含该词)

词“进球”与文本类别的互信息 $MI(X; Y) = \sum_{x \in \{1, 2\}} \sum_{y \in \{0, 1\}} P(X = x, Y = y) \log_2 \frac{P(X=x, Y=y)}{P(X=x)P(Y=y)}$.

• 首先, 计算边缘概率

- $P(X = 1) = \frac{30}{50} = 0.6, P(X = 2) = \frac{20}{50} = 0.4$
- $P(Y = 1) = \frac{26}{50} = 0.52, P(Y = 0) = \frac{24}{50} = 0.48$

• 其次, 计算联合概率

- $P(X = 1, Y = 1) = \frac{25}{50} = 0.5, P(X = 1, Y = 0) = \frac{5}{50} = 0.1$
- $P(X = 2, Y = 1) = \frac{1}{50} = 0.02, P(X = 2, Y = 0) = \frac{19}{50} = 0.38$

• 代入公式

- $MI(X; Y) = 0.5 \log_2 \frac{0.5}{0.6 \times 0.52} + 0.1 \log_2 \frac{0.1}{0.6 \times 0.48} + 0.02 \log_2 \frac{0.02}{0.4 \times 0.52} + 0.38 \log_2 \frac{0.38}{0.4 \times 0.48} \approx 0.46$

统计项	体育类	科技类	总计
包含“进球”文本数	25	1	26
不包含“进球”文本数	5	19	24
包含“算法”文本数	2	18	20
不包含“算法”文本数	28	2	30

In-class Practice II: CBOW

- 我们知道连续词袋模型 (Continuous Bag-of-Words Model, CBOW) 的目标是: 用上下文词的平均向量预测目标词。请写出它的目标函数, 假设上下文窗口是 $[-l, l]$ 。
若是让你训练这个模型, 你觉得会有什么挑战, 如何处理呢?

Ans. to In-class Practice III: CBOW

- **连续词袋模型** (Continuous Bag-of-Words Model, CBOW):

$$J = -\frac{1}{N} \sum_{i=1}^N \log P(w_t | w_{t-l}, w_{t-l+1}, \dots, w_{t-1}, w_{t+1}, \dots, w_{t+l-1}, w_{t+l})$$

- 其中 $P(w_t | context(w_t)) = \frac{\exp(u_c^\top u_t)}{\sum_{i=1}^V \exp(u_i^\top u_c)}$
 - u_c : 上下文词向量的平均向量 ($\frac{1}{2l} \sum_{w \in context(w_t)} u_w$)

- 可能问题
 - 训练词典太大，需要采用负采样